

Equity Markets Positioning Model

Flows Deteriorate, Positioning Holds Steady as Conviction Fades

CITI'S TAKE

Investor conviction continues to fade across global equity markets as flows weaken and risk appetite moderates. US positioning remains modestly bullish, but declining gross exposure points to increasing caution beneath the surface. Europe has experienced a more pronounced deterioration in flows, although positioning remains broadly stable, while bearish exposure continues to accumulate across Asian markets, particularly in the KOSPI and Nikkei. The dominant theme is the growing disconnect between weakening flows and relatively resilient positioning. As a result, positioning conditions remain insufficiently stretched to support a broad capitulation scenario, with localized short squeezes and position unwinds representing the more significant near-term risks.

US – Positioning rebuild loses momentum as shorts return — Recent US positioning trends have softened as renewed short activity offset otherwise limited investor demand. S&P 500 flows turned modestly bearish last week, while longer-term positioning remains only slightly above neutral and well below June extremes. More notable is the continued decline in gross exposure, suggesting investors are increasingly disengaged ahead of key inflation data. While unrealized losses have increased across both long and short books, overall exposure remains moderate. Consequently, P&L stress does not yet translate into the type of exposure extremes typically associated with broad capitulation events. Small caps remain the notable exception, where concentrated long exposure and elevated unrealized losses increase the risk of position unwinds should market weakness persist.

Europe – Selling pressure rises, but positioning remains resilient – Equity flows deteriorated materially last week, driven primarily by long liquidation in EuroStoxx and aggressive short additions in DAX. Despite weaker flow dynamics, aggregate positioning across EuroStoxx, FTSE and European Banks remain broadly stable due to offsetting investor activity. The most notable setup remains the DAX, where an unusually large short base has developed. A large concentration of short positions here remains loss-making, with average entry levels materially below current prices. As a result, the market remains vulnerable to renewed short covering should economic headwinds ease.

Asia – Shorts exposure rebuilds – Positioning across Nikkei and KOSPI continued to weaken as investors added shorts and reduced longs across both indices. Both markets are increasingly characterized by one-sided bearish positioning, though current profits on the short book remain relatively modest. This leaves flow asymmetry skewed toward potential short covering should markets rebound. Elsewhere, Hang Seng and China A50 positioning remains near neutral, although recent flows have diverged, with Hang Seng exhibiting signs of renewed weakness. The growing contrast between increasingly shorted KOSPI positioning and constructive Nasdaq exposure is particularly notable. This divergence suggests that investors are differentiating between regional semiconductor exposures rather than expressing a uniform view on the broader AI thematic.

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Current Themes

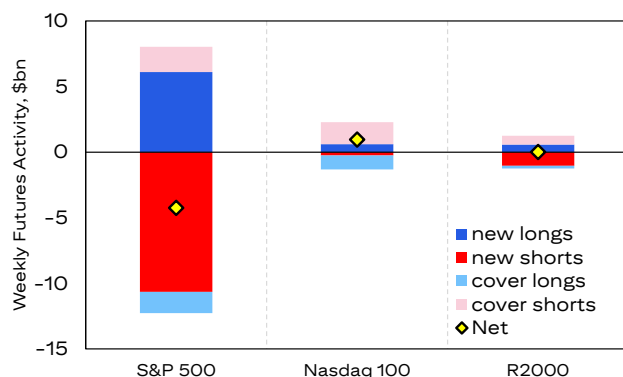
US equities closed higher last week, albeit marginally in what has been a volatile week driven by a mix of geopolitics (re-escalation of US/Iran hostilities) and a strong payrolls print ([US Economics - "Stable" labor market means later rate cuts](#)), which saw the market decline on the day. The case of 'good data is bad news for markets' defines the recent payrolls release and it leaves a potential policy dilemma for the Fed on rate setting. Our economists now believe the rate decision will be strongly linked to this week's CPI release. Anything close to 0.4% MoM is likely to provoke a hike. All this is taking place as investors navigate a complex backdrop of synchronized hawkish central bank views across regions.

Positioning across US equities has been relatively muted recently, with mixed flows and limited net positioning changes. For the S&P 500, new short positioning represented a larger part of last week's flows (Figure 1). There was less to discern across the other indices from a flow perspective.

Longer term normalised trends over the month have been pointing towards a marginal increase in levels for the S&P (+0.5) and Nasdaq (+0.9). (Figure 2), but distant from the euphoric levels seen in early June.

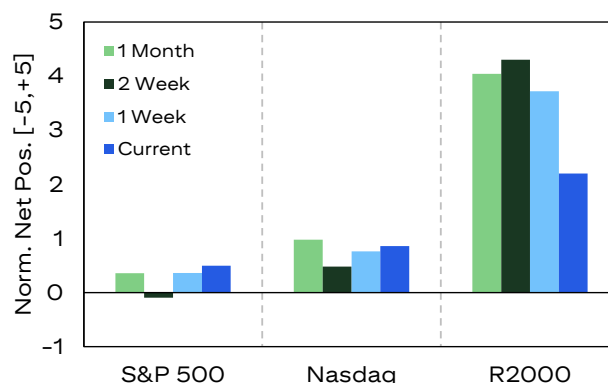
Despite positioning levels appearing supportive for the S&P, gross positioning has compressed considerably (Figure 3) suggesting a degree of disengagement, less surprising given current yields and the upcoming potential CPI risk event.

Figure 1. Bearish flow interrupts US positioning recovery.



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Source: Citi Research

Figure 2. US Large Cap positioning momentum eases.

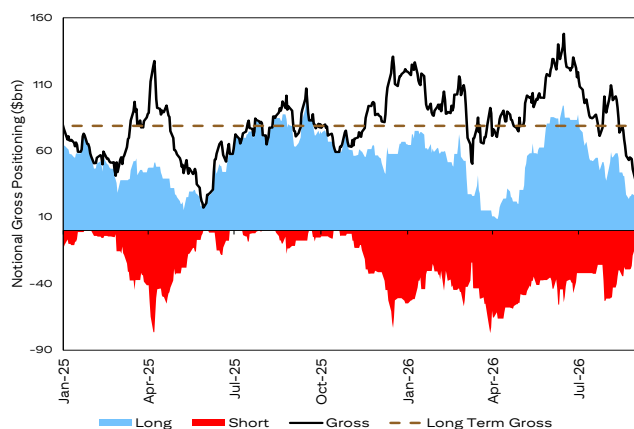


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Recent volatility has driven a deterioration in positioning P&L, with both long and short books experiencing elevated unrealized losses. In both the S&P 500 and Nasdaq, on aggregate half of long and short positions are currently in loss. An upside CPI surprise would likely boost short-book profitability and reinforce the bearish positioning that has emerged through the recent build-up in short exposure.

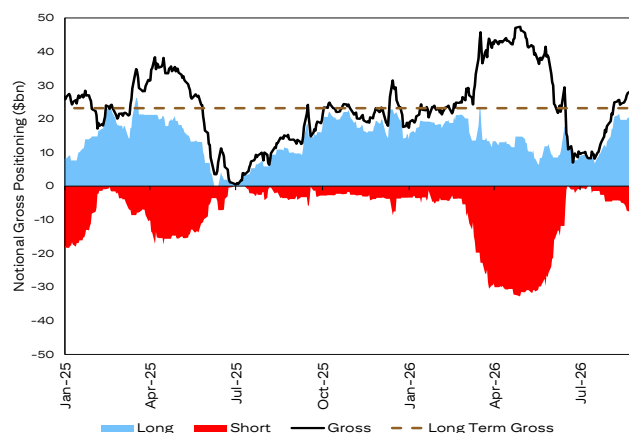
While overall losses are of concern, taken in conjunction with current positioning levels, capitulation on either side is less likely to lead to considerable pressure on the index. The risk is greater for US small caps, more elevated long concentration and with 93% of positions offside leaves the index still susceptible to profit-taking unwinds if bearish momentum accelerates.

Figure 3. Investor conviction fades as S&P 500 gross exposure falls.



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Figure 4. EuroStoxx gross levels holding above long term levels despite recent market weakness.



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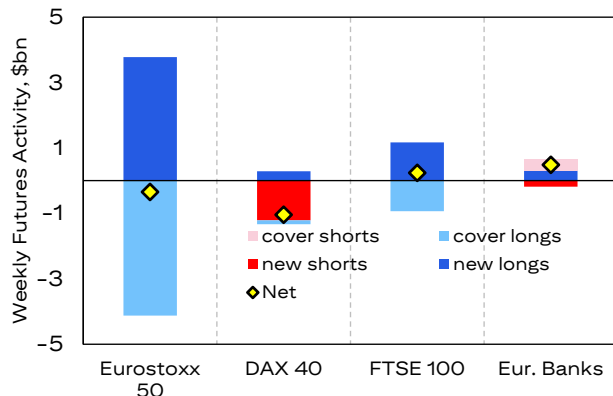
Europe

European markets witnessed near consensus declines across regional indices, with EuroStoxx, DAX and even FTSE edging lower over the week. The recent escalation (US/Iran) and consequential step up in energy prices formed the main driver behind recent alongside a step-up in yields.

For EuroStoxx, net selling was driven entirely by long liquidation, while DAX saw the most aggressive European short addition. FTSE attracted new risk flows, but this was offset a similar amount of long unwinds over the week. European Banks was the only index to see more decisive positive flows from short unwinds and new risk.

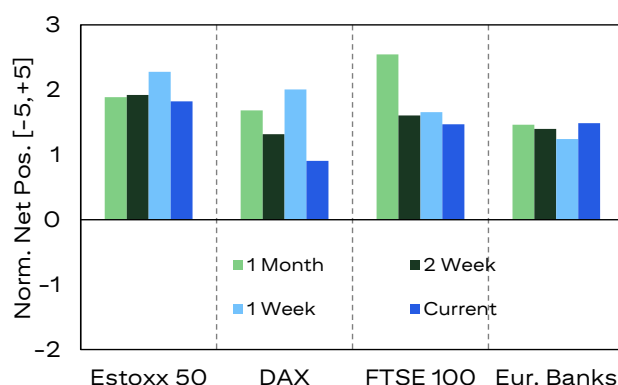
Due to the offsetting nature of recent flows, net positioning across EuroStoxx, FTSE and Euro Banks have remained largely unchanged (Figure 6). For DAX, weakness observed last week, represents an acceleration of recent bearish trends.

Figure 5. Long unwinds drive bearish shift for Europe...



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Figure 6. ... positioning remains intact despite deterioration.



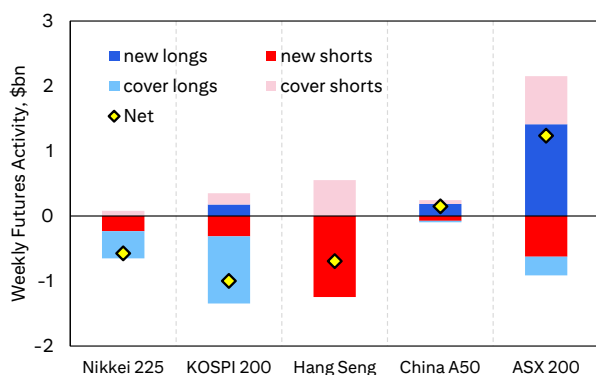
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Positioning risks are elevated for DAX, here P&L (-2.1) loss levels have accelerated, and is currently the worst across indices tracked (Figure 9). The average short entry level is 25,330, and nearly 85% of the short book is currently loss making. Despite the recent index decline, shorts entered earlier in the summer remain deeply in loss and the abnormally large short base could see increased covering supporting the index higher.

Asia/EM

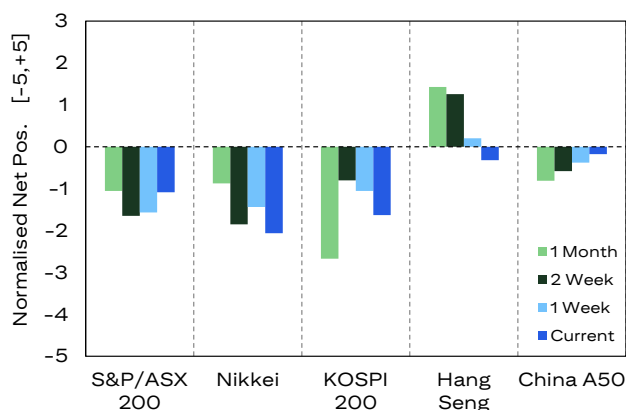
Positioning in Nikkei and KOSPI weakened over the week driven by a mix of long unwinds and new short positioning (Figure 7), with both indices now near one-sided short and witnessing a similar rising trend in bearish positioning (Figure 8). While the short book is largely profitable, profit size is less extensive, and the greater risk will reside on potential forced covering if both markets see a rapid rebound. So far investor flows have remained soft, with the no clear change in trends witnessed despite the tepid rebound in KOSPI and an easing in outflows ([South Korea Economics - A Jump in Foreign Reserves & Forward Position; Weak Capital Outflow Pressure](#)). Both Hang Seng and A50 are near neutral but saw diverging weekly flows with the former trending lower.

Figure 7. Bearish exposure continues to build across Nikkei, KOSPI and Hang Seng.



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Figure 8. Positioning trends largely unchanged across Asia.



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A distinct peculiarity lies within the current KOSPI short vs Nasdaq long positioning setup. Both indices exhibit significant semiconductor exposure, but opposite positioning setup, suggesting that diverging demand drivers are emerging despite both sharing the same 'semiconductor' label.

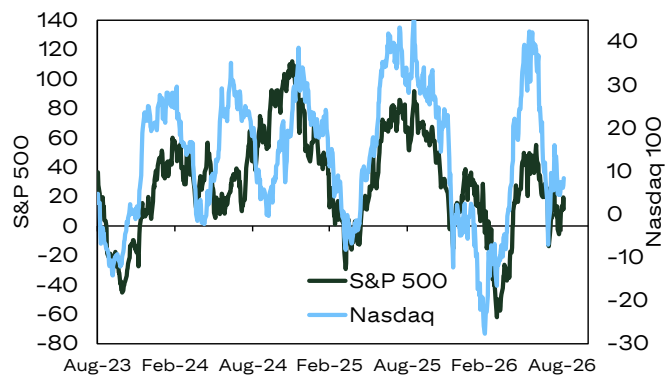
Figure 9. Positioning change across global equities.

	By 1w change	Pos	PnL
Change			
ASX 200	0.5	-1.1	-0.1
Eur Banks	0.2	1.5	0.5
China A50	0.2	-0.2	0.0
S&P Mini	0.1	0.5	0.2
Nasdaq	0.1	0.9	-0.2
FTSE	-0.2	1.5	0.3
EuroStoxx	-0.5	1.8	-0.2
Hang Seng	-0.5	-0.3	0.0
KOSPI 200	-0.6	-1.6	0.1
Nikkei	-0.6	-2.1	0.1
DAX	-1.1	0.9	-2.1
MSCI EM	-1.1	1.5	0.4
DM xUS	-1.2	-1.1	0.1
R2000	-1.5	2.2	-1.1

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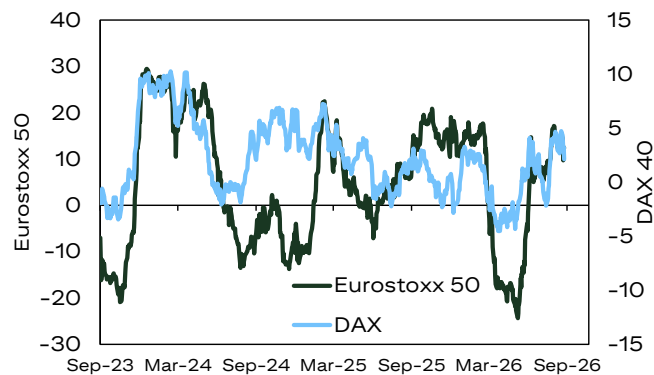
How has Positioning Changed?

Figure 10. S&P 500 (ES) and Nasdaq 100 (NQ) net positioning



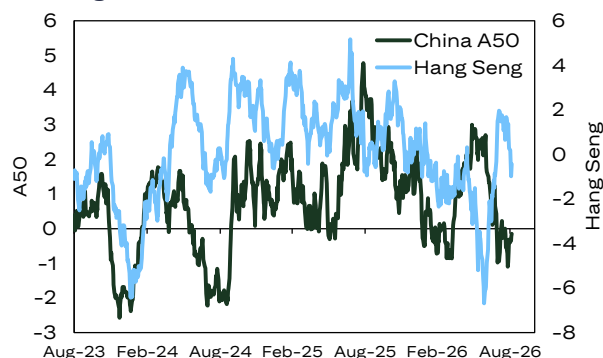
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Figure 11. EuroStoxx 50 (VG) and DAX 40 (GX) net positioning



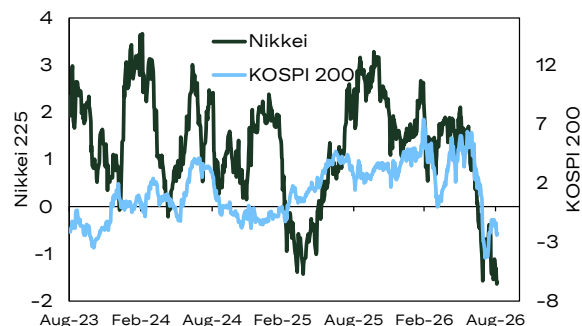
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Figure 12. Hang Seng (HI) and China A50 (XU) net positioning



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Figure 13. Nikkei 225 (NI) and KOSPI 200 (KM) net positioning



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Figure 14. Ranking of Equity Market positioning

04/09/2026	By Position	PnL	By 1w change	Pos	PnL	By PnL	PnL	Pos	By Position	Percentile
R2000	2.2	-1.1	ASX 200	0.5	-1.1	-0.1	Eur Banks	0.5	1.5	79%
EuroStoxx	1.8	-0.2	Eur Banks	0.2	1.5	0.5	MSCI EM	0.4	1.5	75%
MSCI EM	1.5	0.4	China A50	0.2	-0.2	0.0	FTSE	0.3	1.5	70%
Eur Banks	1.5	0.5	S&P Mini	0.1	0.5	0.2	S&P Mini	0.2	0.5	64%
FTSE	1.5	0.3	Nasdaq	0.1	0.9	-0.2	KOSPI 200	0.1	-1.6	63%
DAX	0.9	-2.1	FTSE	-0.2	1.5	0.3	Nikkei	0.1	-2.1	52%
Nasdaq	0.9	-0.2	EuroStoxx	-0.5	1.8	-0.2	DM xUS	0.1	-1.1	39%
S&P Mini	0.5	0.2	Hang Seng	-0.5	-0.3	0.0	China A50	0.0	-0.2	37%
China A50	-0.2	0.0	KOSPI 200	-0.6	-1.6	0.1	Hang Seng	0.0	-0.3	27%
Hang Seng	-0.3	0.0	Nikkei	-0.6	-2.1	0.1	ASX 200	-0.1	-1.1	27%
DM xUS	-1.1	0.1	DAX	-1.1	0.9	-2.1	Nasdaq	-0.2	0.9	27%
ASX 200	-1.1	-0.1	MSCI EM	-1.1	1.5	0.4	EuroStoxx	-0.2	1.8	23%
KOSPI 200	-1.6	0.1	DM xUS	-1.2	-1.1	0.1	R2000	-1.1	2.2	4%
Nikkei	-2.1	0.1	R2000	-1.5	2.2	-1.1	DAX	-2.1	0.9	0%

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Figure 15. Summary

		S&P 500	Nasdaq	R2000	Estox 50	DAX	FTSE 100	Eur. Banks	S&P/ASX 200	Nikkei	MSCI EAFE	MSCI EM	KOSPI 200	Hang Seng	China A50
(\$bn)	04/09/2026	ES	NQ	RTY	VG	GX	Z	CA	XP	NI	MFS	MES	KM	HI	XU
Equity	Net Notional	11.7	8.4	5.4	12.4	2.1	4.3	1.9	-2.1	-1.6	-1.7	3.5	-2.2	-0.4	-0.1
	Percentile	27%	39%	79%	63%	52%	75%	70%	27%	0%	27%	64%	4.3%	37%	23%
Shorts	Notional	-20.1	-6.8	-3.2	-8.5	-8.2	-1.6	0.0	-4.5	-2.1	-4.0	-0.1	-2.5	-2.8	-0.3
	Percentile	45%	55%	55%	58%	97%	57%	5%	80%	92%	71%	6%	90%	83%	32%
	Avg. Entry	7,721	29,551	2,979	6,385	25,327	10,844	--	9,014	65,329	3,252.0	1,640	1,066.3	25,356	14,679
	% offside	66%	60%	33%	49%	84%	33%	--	56%	11%	52%	100%	6%	100%	49%
Longs	Notional	31.8	15.2	8.6	20.9	10.4	5.9	1.9	2.5	0.5	2.2	3.6	0.2	2.4	0.2
	Percentile	19%	34%	85%	79%	97%	80%	51%	48%	0%	36%	35%	6%	63%	14%
	Avg. Entry	7,696	29,722	3,005	6,414	25,843	10,793	316.4	9,028	66,517	3,248	1,704	1,053.9	25,333.3	14,628
	% offside	46%	63%	93%	53%	37%	49%	30%	47%	100%	33%	0%	100%	53%	41%
	Ref Price	7,730	29,565	2,969	6,397	26,086	10,841	322.8	9,015	64,668	3,250	1,740	1,048.6	25,619.3	14,672
Activity	1-day	7.8	-1.7	0.2	-0.9	-0.0	-0.8	0.2	0.1	-0.0	0.0	0.0	0.2	0.4	0.1
	Type	shorts	shorts		shorts	shorts	shorts	longs		shorts		longs			
	1-week	-4.2	1.0	0.0	-0.3	-1.0	0.2	0.5	1.2	-0.6	0.3	-0.9	-1.0	-0.7	0.1
	Normalised Scores (-5/+5)														
Positioning	Equity implied	0.5	0.9	2.2	1.8	0.9	1.5	1.5	-1.1	-2.1	-1.1	1.5	-1.6	-0.3	-0.2
	ETF	1.3	0.6		-0.5	-1.1	0.4	1.4		1.2		1.3			
	3m Net Pos	0.9	0.7	2.2	0.6	-0.1	0.9	1.4	-1.1	-0.4	-1.1	1.4	-1.6	-0.3	-0.2
	1w Change	-0.2	-1.5	-1.5	-0.2	-0.6	0.2	0.2	0.5	-0.3	-1.2	-0.5	-0.6	-0.5	0.2
P&L	Equity implied	0.2	-0.2	-1.1	-0.2	-2.1	0.3	0.5	-0.1	0.1	0.1	0.4	0.1	-0.0	0.0
	ETF	0.4	-0.3		-0.6	-0.4	0.9	1.6		1.1		1.0			
	3m Net PnL	0.3	-0.3	-1.1	-0.4	-1.2	0.6	1.0	-0.1	0.6	0.1	0.7	0.1	-0.0	0.0
	1w Change	-0.0	-0.0	-0.9	-0.2	-1.0	0.4	0.5	0.0	0.3	-1.0	0.3	0.2	-0.1	0.1

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Figure 16. Average Entry Price Levels and Profit Breakeven Levels

		S&P 500	Nasdaq	R2000	Estoxx 50	DAX	FTSE 100	Eur. Banks	S&P/ASX 200	Nikkei	MSCI EAFE	MSCI EM	KOSPI 200	Hang Seng	China A50
	04/09/2026	ES	NQ	RTY	VG	GX	Z	CA	XP	NI	MFS	MES	KM	HI	XU
Notional	\$bn	11.7	8.4	5.4	12.4	2.1	4.3	1.9	-2.1	-1.6	-1.7	3.5	-2.2	-0.4	-0.1
	(-5/+5)	0.5	0.9	2.2	1.8	0.9	1.5	1.5	-1.1	-2.1	-1.1	1.5	-1.6	-0.3	-0.2
PnL	\$bn	0.1	-0.1	-0.1	-0.1	-0.2	0.0	0.0	-0.0	0.0	0.0	0.1	0.0	-0.0	0.0
	(-5/+5)	0.2	-0.2	-1.1	-0.2	-2.1	0.3	0.5	-0.1	0.1	0.1	0.4	0.1	-0.0	0.0
Ref Price		7,730	29,565	2,969	6,397	26,086	10,841	322.8	9,015	64,668	3,250	1,740	1,049	25,619	14,672
Shorts	Size (\$bn)	-20.1	-6.8	-3.2	-8.5	-8.2	-1.6	0.0	-4.5	-2.1	-4.0	-0.1	-2.5	-2.8	-0.3
	Avg Price	7,721	29,551	2,979	6,385	25,327	10,844	--	9,014	65,329	3,252	1,640	1,066.3	25,356	14,679
	Avg P&L	-0.1%	0.0%	0.3%	-0.2%	-2.9%	0.0%	--	0.0%	1.0%	0.1%	-5.8%	1.7%	-1.0%	0.0%
	Offside (%)	66%	60%	33%	49%	84%	33%	--	56%	11%	52%	100%	6%	100%	49%
- Offside price															
	100%	7,788	29,702	3,010	6,568	26,472	10,858	--	9,153	67,778	3,268	1,671	1,119.3	25,475	14,729
	75%	7,757	29,647	2,990	6,461	25,363	10,853	--	9,021	65,666	3,262	1,631	1,070.8	25,439	14,680
	50%	7,715	29,486	2,972	6,308	25,089	10,843	--	9,002	64,930	3,246	1,631	1,055.6	25,326	14,631
	25%	7,684	29,377	2,958	6,276	24,967	10,815	--	8,946	64,688	3,241	1,631	1,048.6	25,252	14,627
	0%	7,454	29,375	2,958	6,258	24,873	10,788	--	8,917	64,648	3,231	1,631	983.8	25,130	14,627
Longs	Size (\$bn)	31.8	15.2	8.6	20.9	10.4	5.9	1.9	2.5	0.5	2.2	3.6	0.2	2.4	0.2
	Avg Price	7,696	29,722	3,005	6,414	25,843	10,793	316.4	9,028	66,517	3,248	1,704	1,053.9	25,333	14,628
	Avg P&L	0.4%	-0.5%	-1.2%	-0.3%	0.9%	0.4%	2.0%	-0.1%	-2.8%	0.1%	2.1%	-0.5%	1.1%	0.3%
	Offside (%)	46%	63%	93%	53%	37%	49%	30%	47%	100%	33%	0%	100%	53%	41%
- Offside price															
	100%	7,600	28,697	2,948	6,258	25,171	10,695	302.4	8,990	65,818	3,186	1,687	1,048.6	23,497	14,598
	75%	7,600	29,459	2,991	6,331	25,539	10,700	308.5	8,990	65,818	3,244	1,698	1,048.6	25,379	14,598
	50%	7,716	29,642	2,999	6,397	25,670	10,821	318.7	8,990	65,819	3,246	1,699	1,048.6	25,589	14,598
	25%	7,732	29,867	3,017	6,490	26,321	10,845	323.2	9,042	66,361	3,263	1,706	1,051.2	25,626	14,627
	0%	7,782	30,116	3,041	6,533	26,571	10,902	326.7	9,122	68,995	3,271	1,725	1,066.6	25,758	14,672

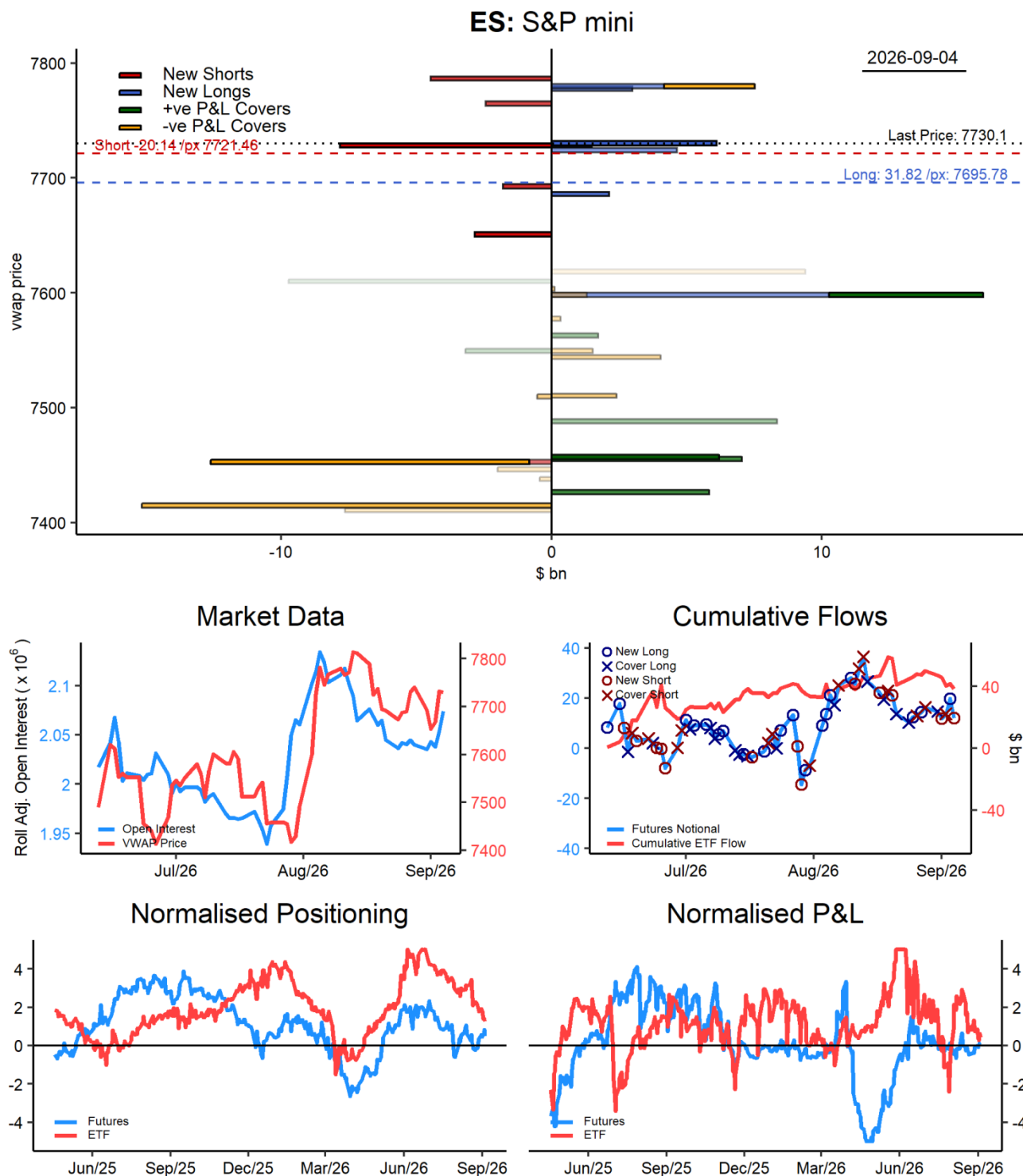
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Equity Markets Positioning Charts

S&P 500

Figure 17. S&P 500 mini contract (ES) - position snapshot and time-series

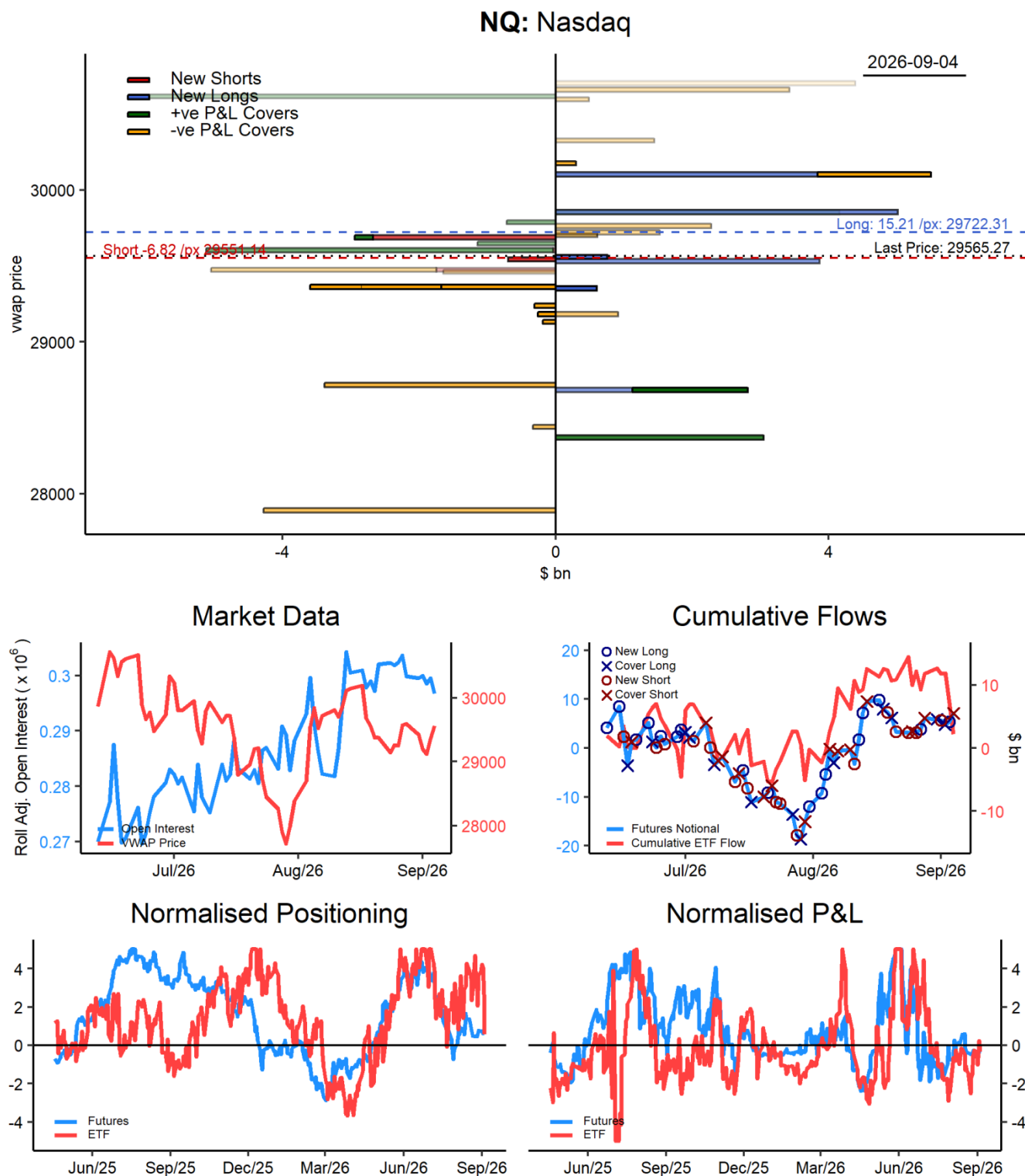


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NASDAQ 100

Figure 18. Nasdaq 100 contracts (NQ) - position snapshot and time-series

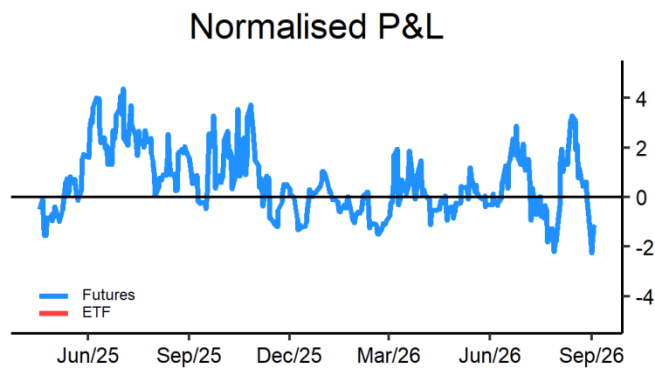
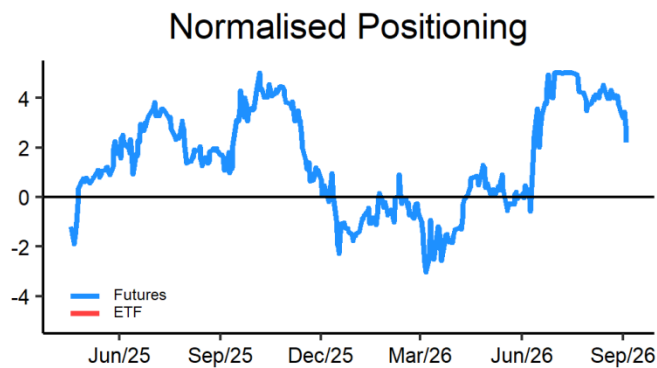
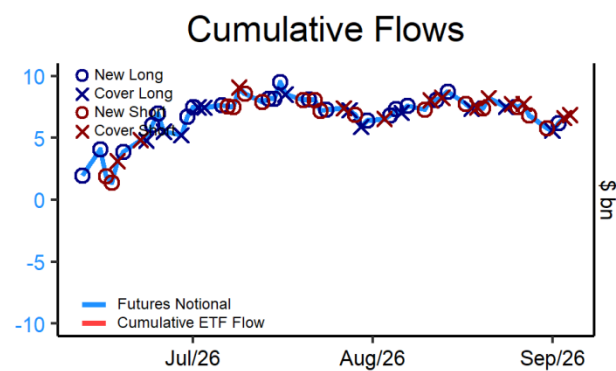
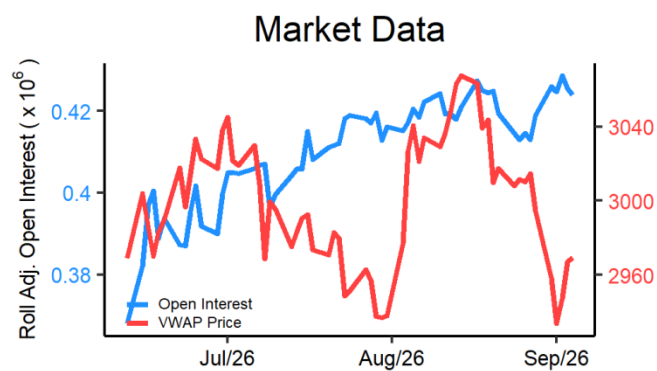
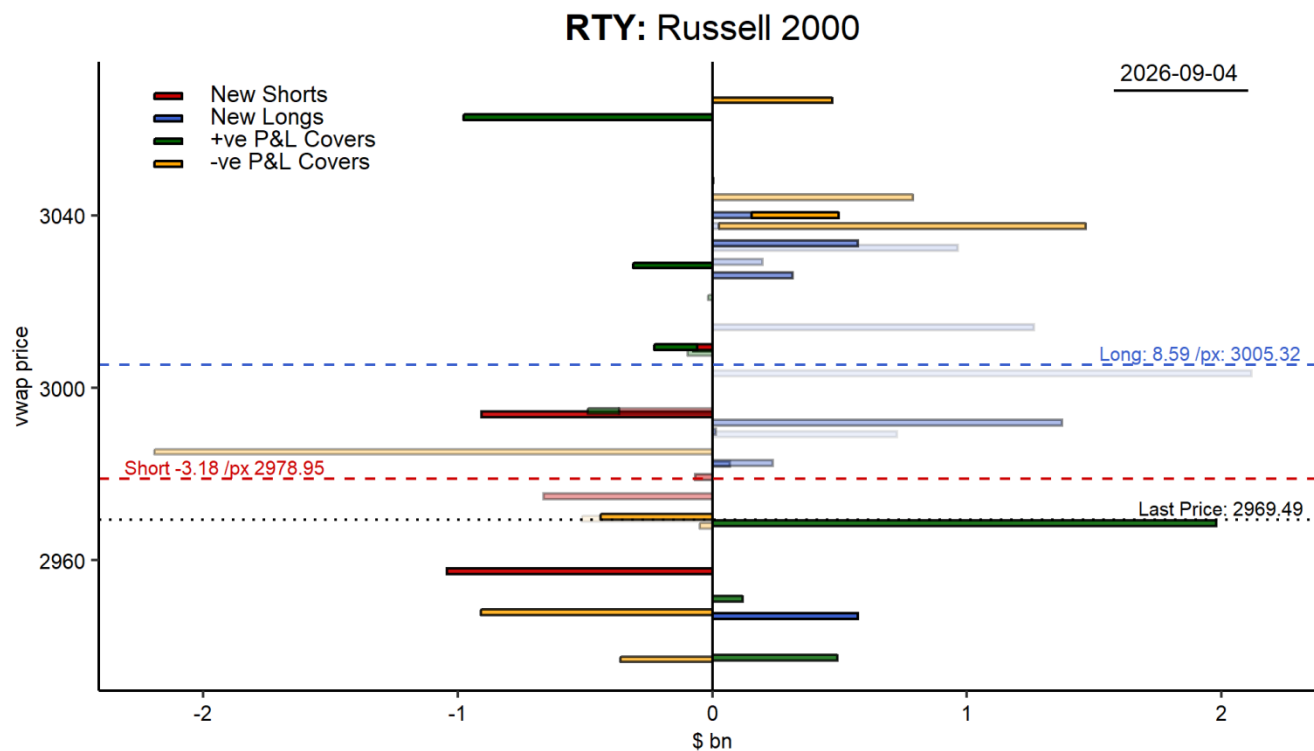


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Source: Citi Research, Bloomberg

Russell 2000

Figure 19. Russell 2000 contracts (RTY) - position snapshot and time-series

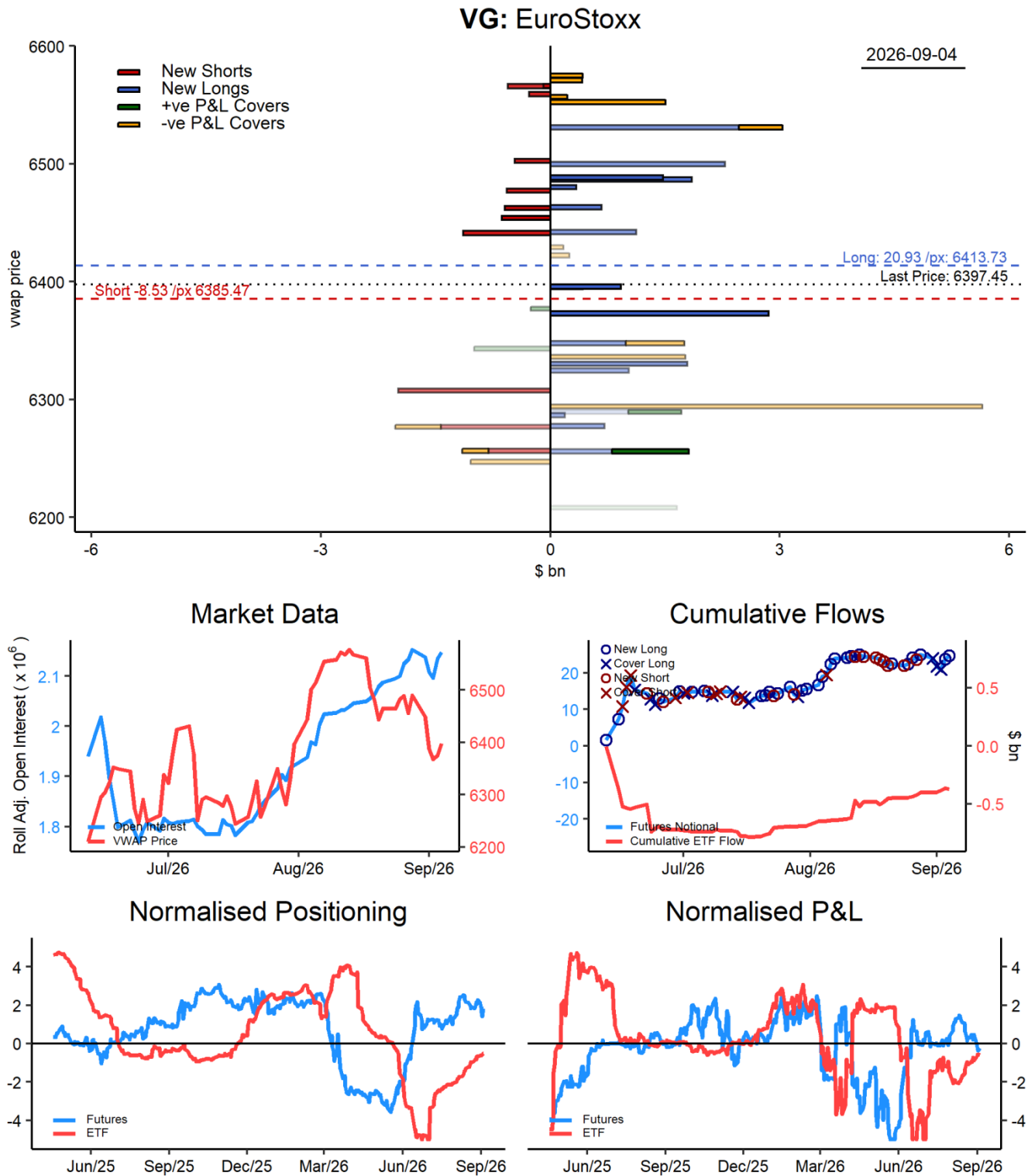


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Source: Citi Research, Bloomberg

EuroStoxx 50

Figure 20. Eurostoxx 50 contracts (VG) - position snapshot and time-series

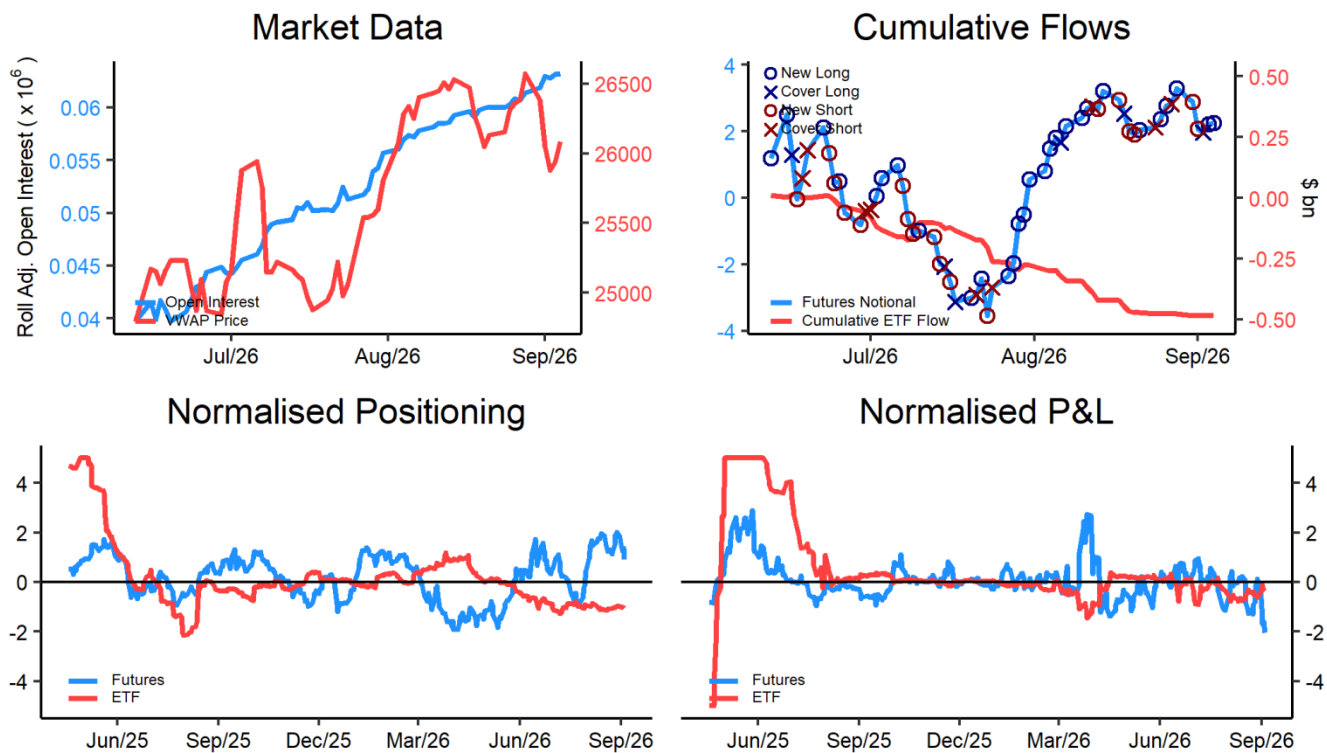
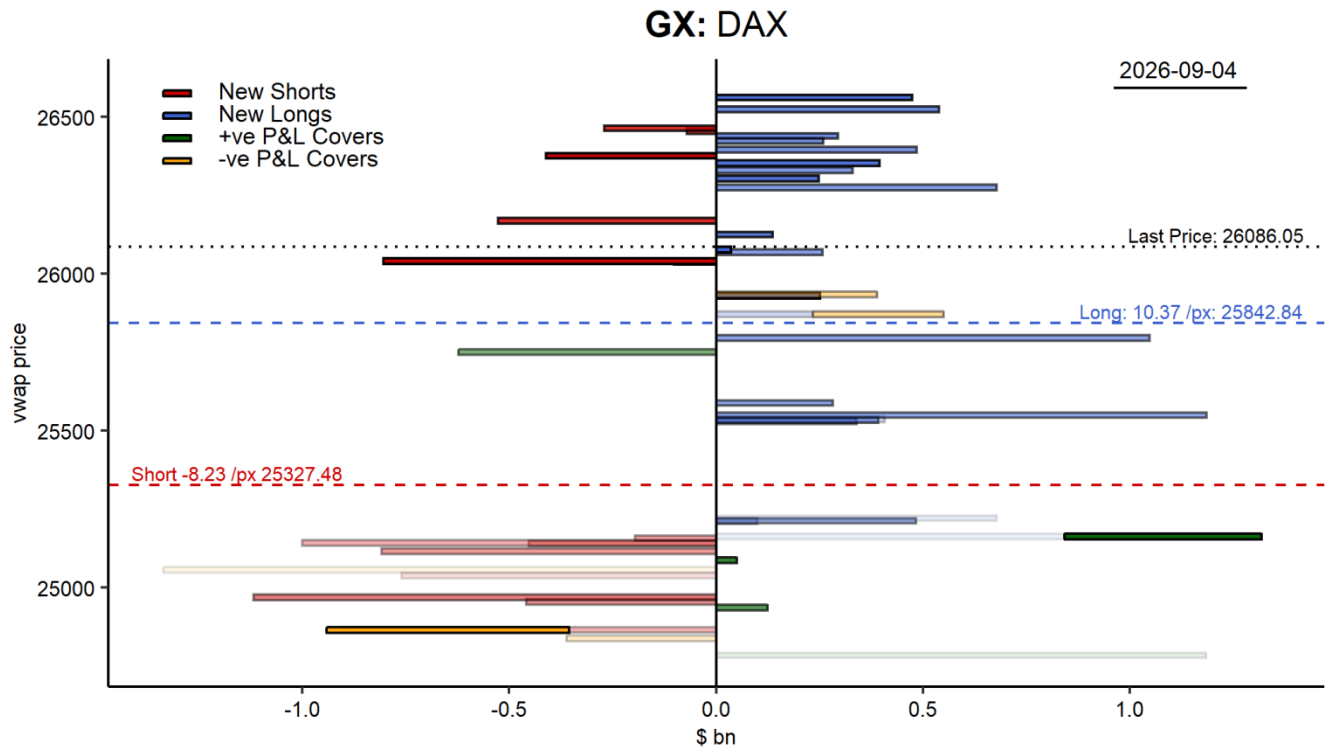


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Source: Citi Research, Bloomberg

DAX

Figure 21. DAX contracts (GX) - position snapshot and time-series

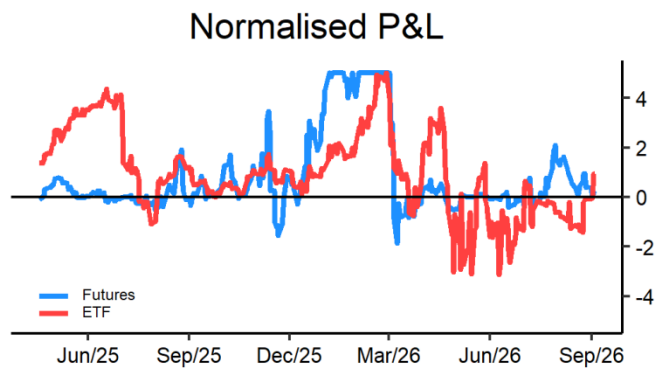
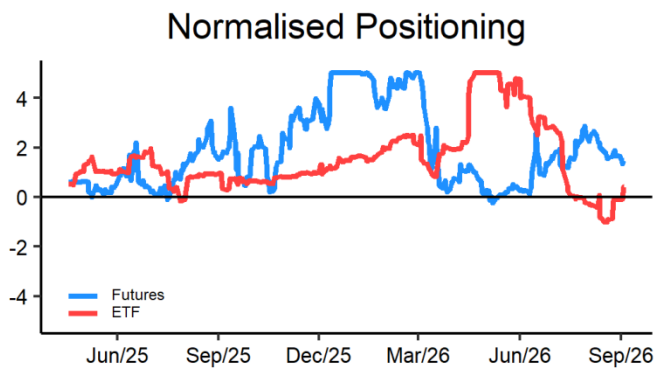
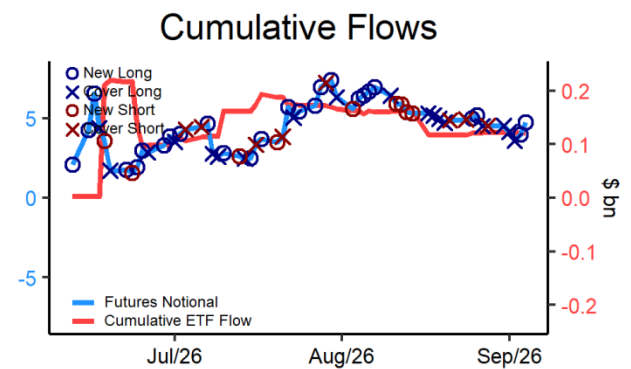
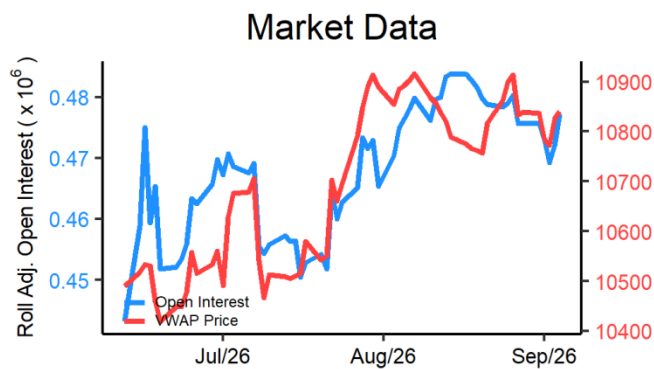
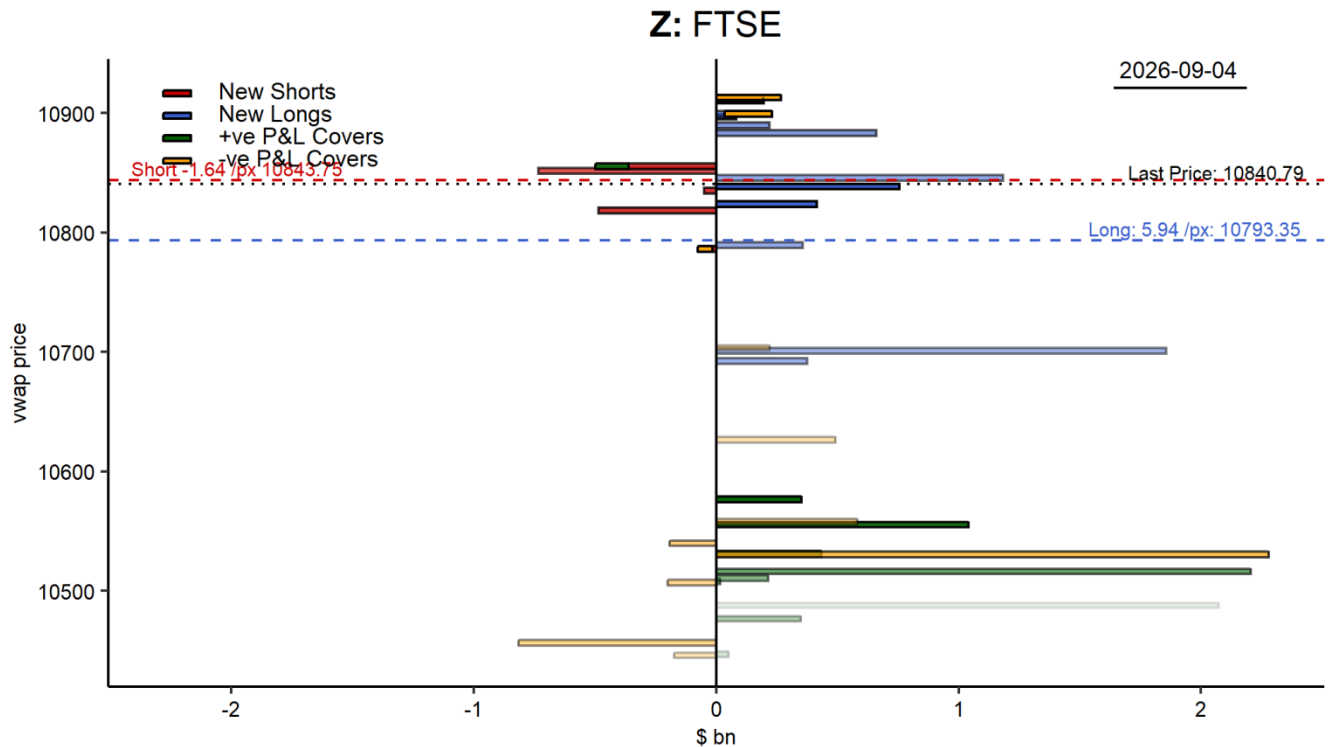


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Source: Citi Research, Bloomberg

FTSE 100

Figure 22. FTSE 100 contracts (Z) - position snapshot and time-series

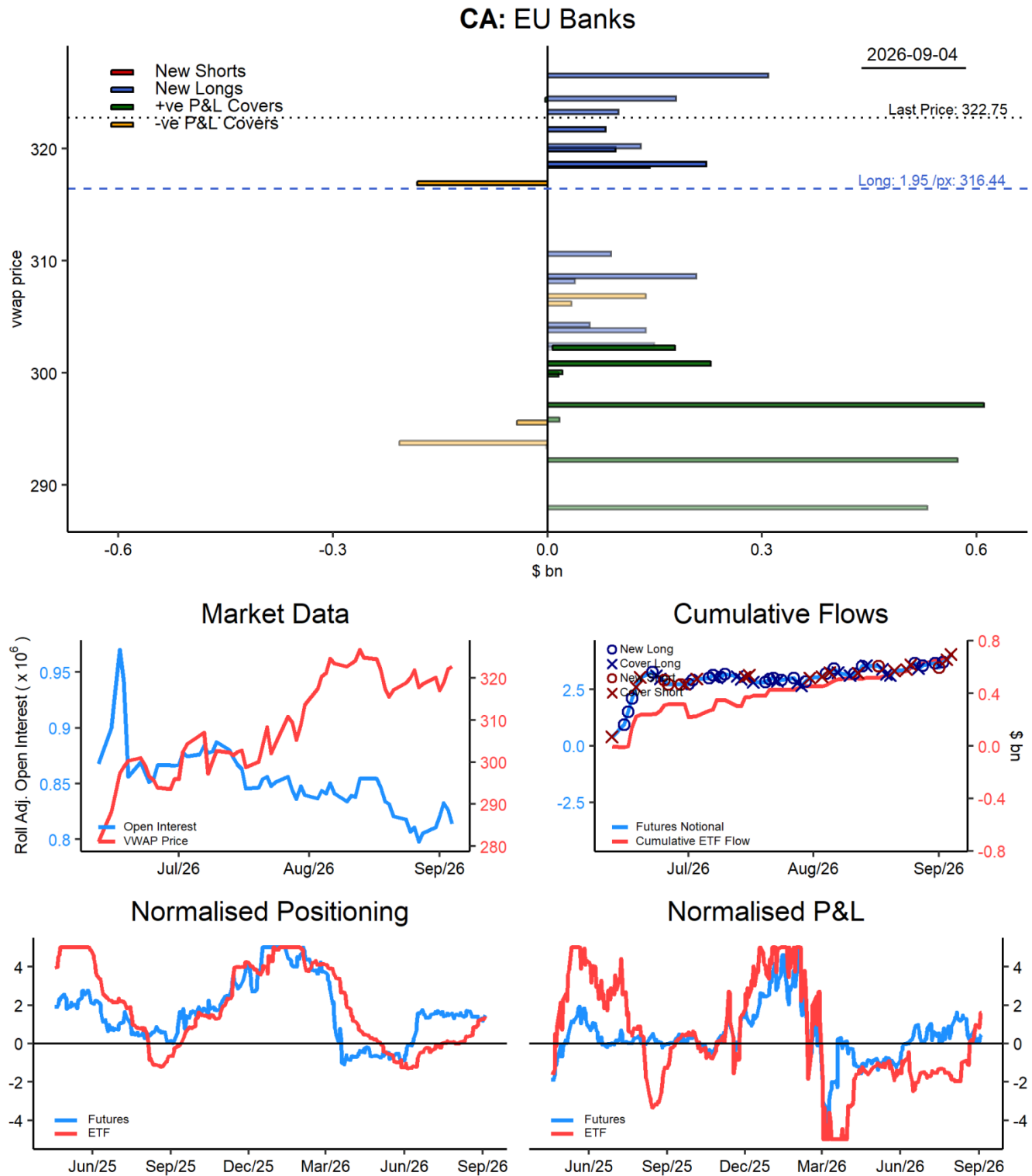


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Source: Citi Research, Bloomberg

European Banks (SX7E)

Figure 23. Eurostoxx Banks (SX7E) Equity contracts (CA) - position snapshot and time-series

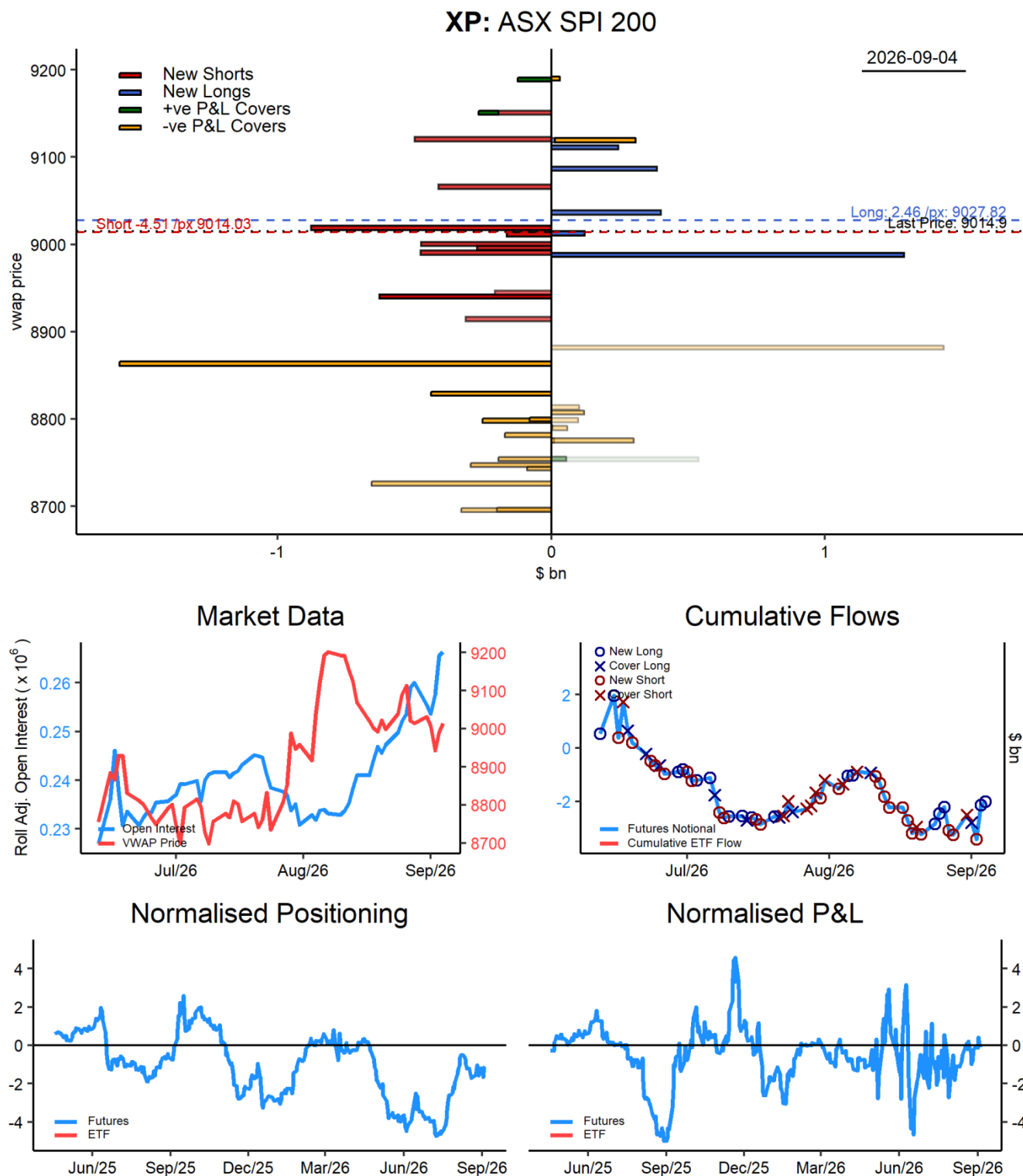


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Source: Citi Research, Bloomberg

S&P ASX 200

Figure 24. S&P ASX 200 Equity contracts (XP) - position snapshot and time-series

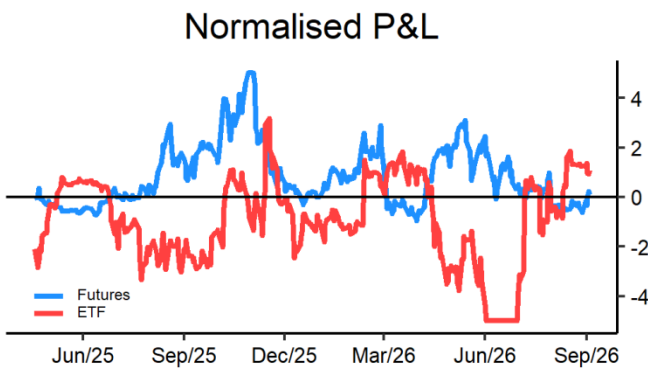
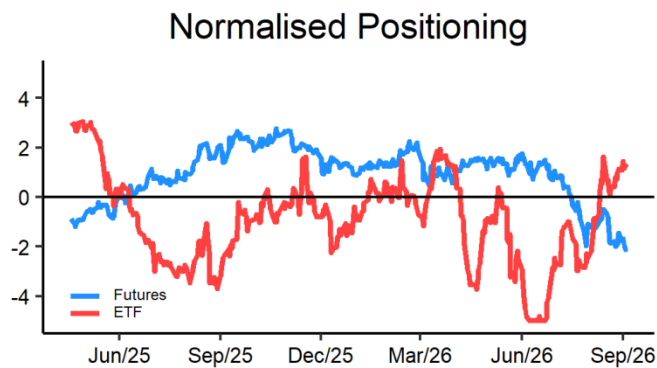
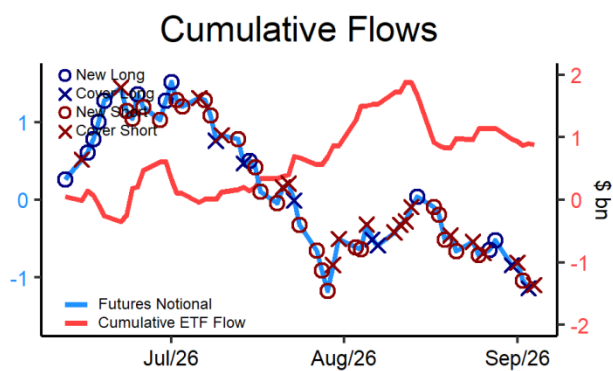
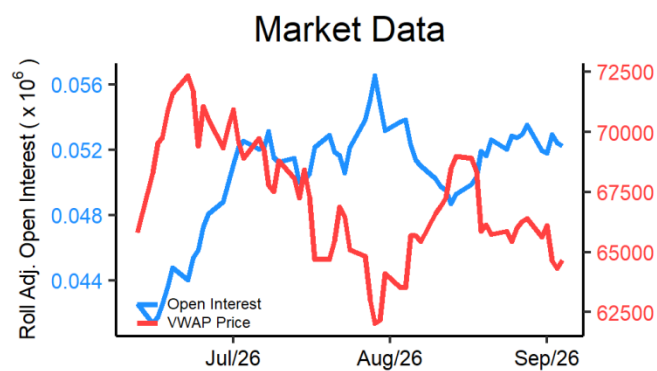
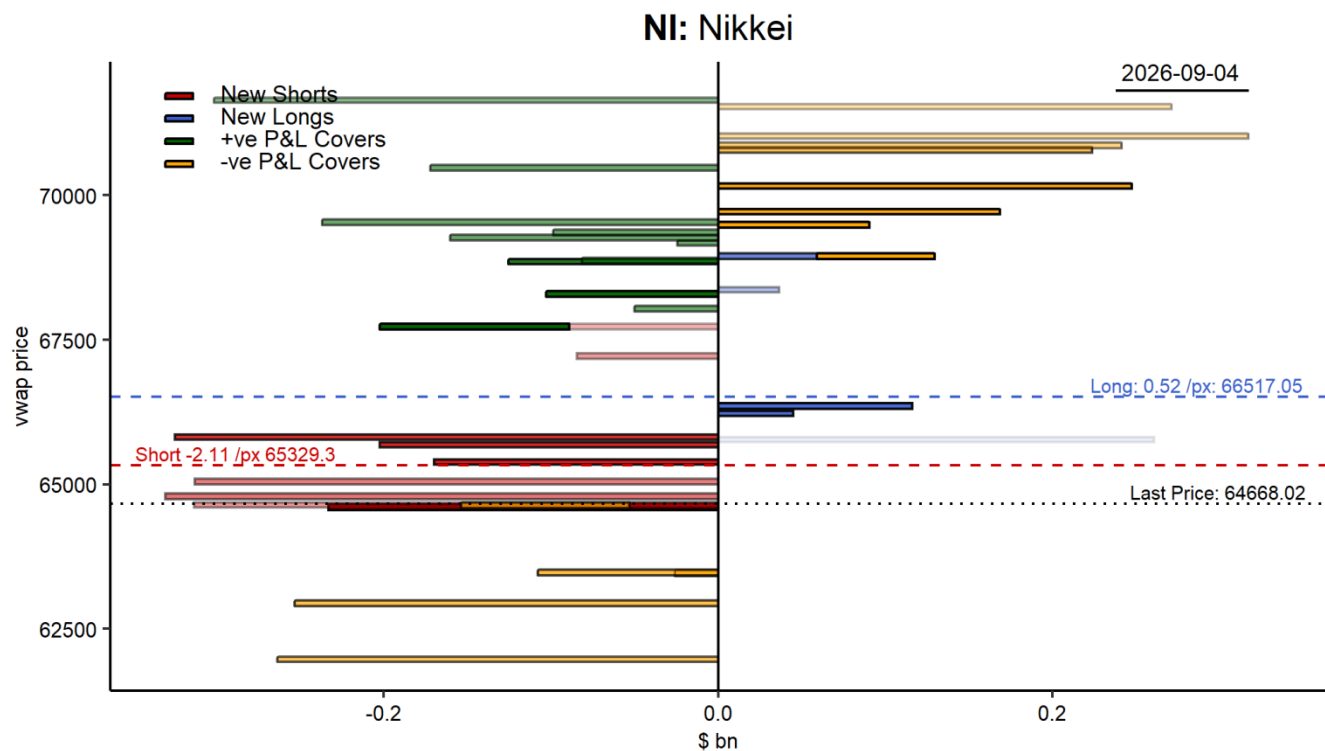


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Source: Citi Research, Bloomberg

Nikkei 225

Figure 25. Nikkei contracts (NI) - position snapshot and time-series

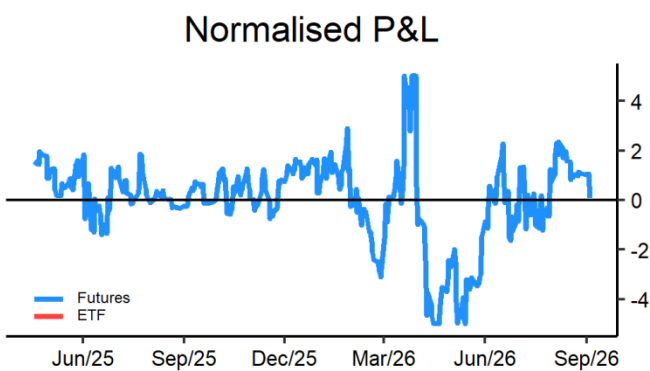
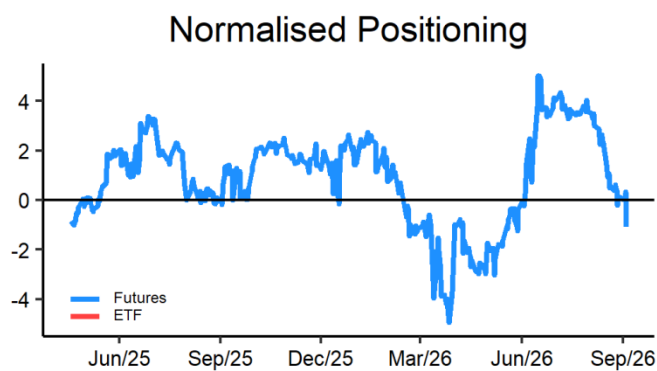
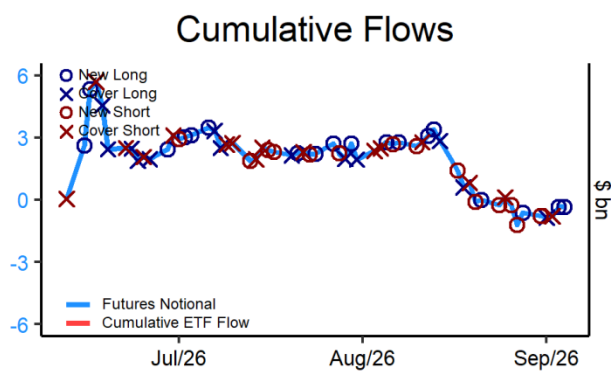
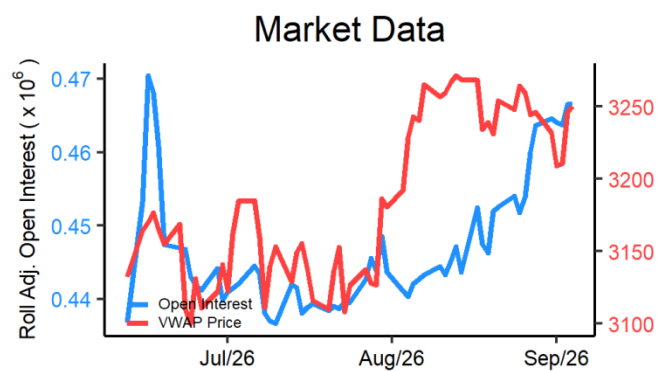
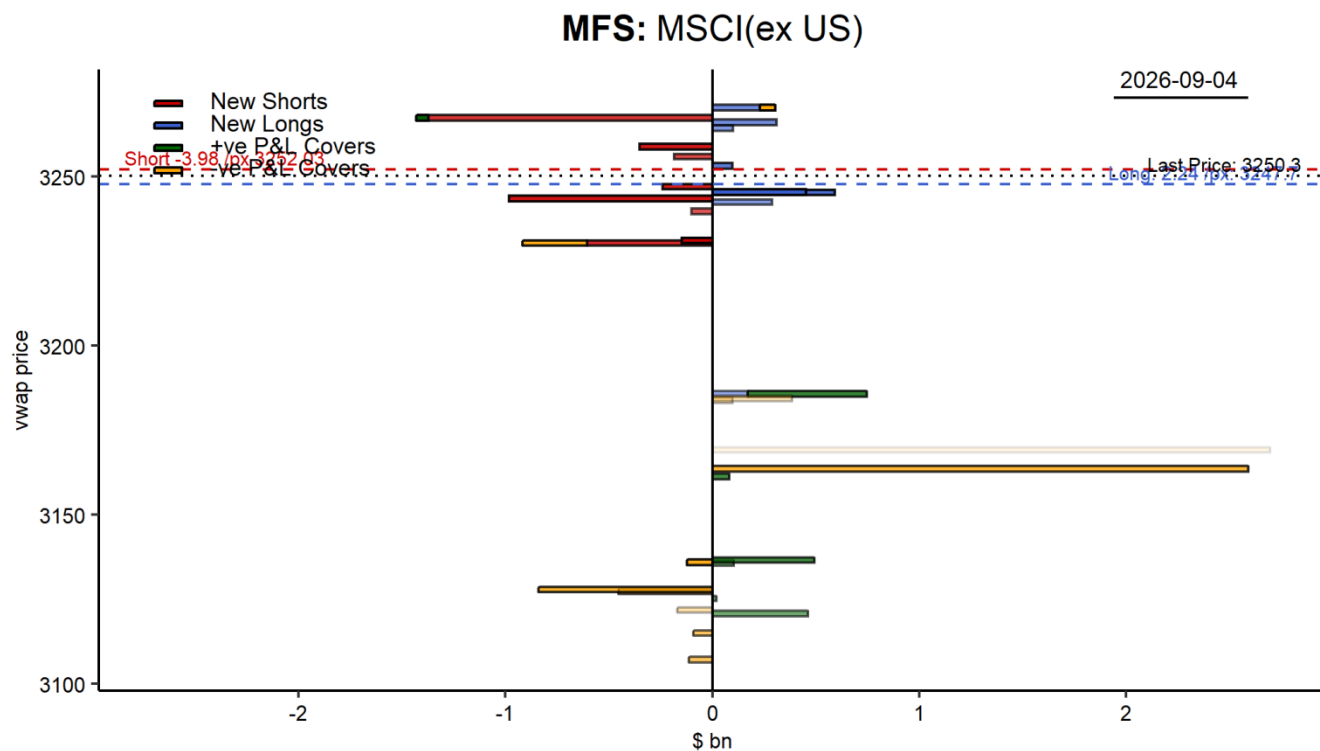


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Source: Citi Research, Bloomberg

MSCI EAFE

Figure 26. MSCI EAFE (DM ex-US) (MFS) - position snapshot and time-series

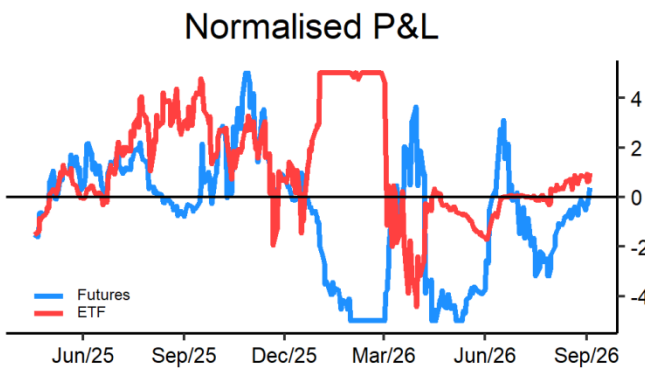
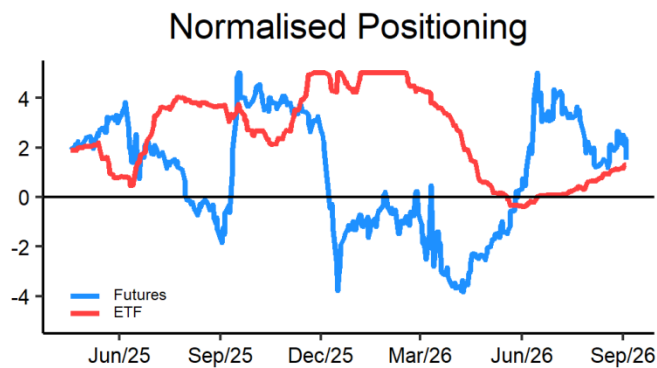
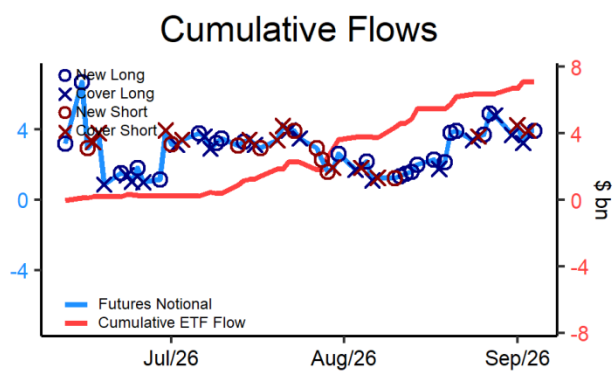
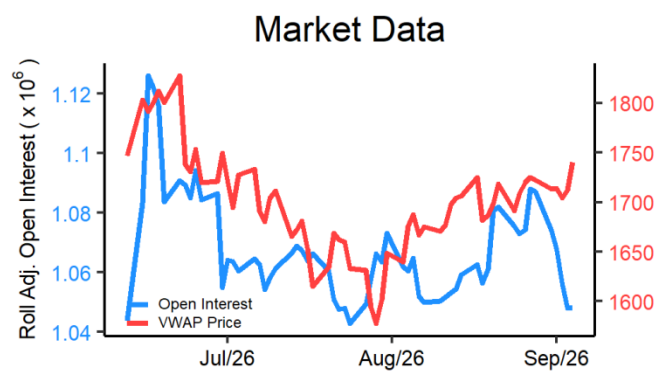
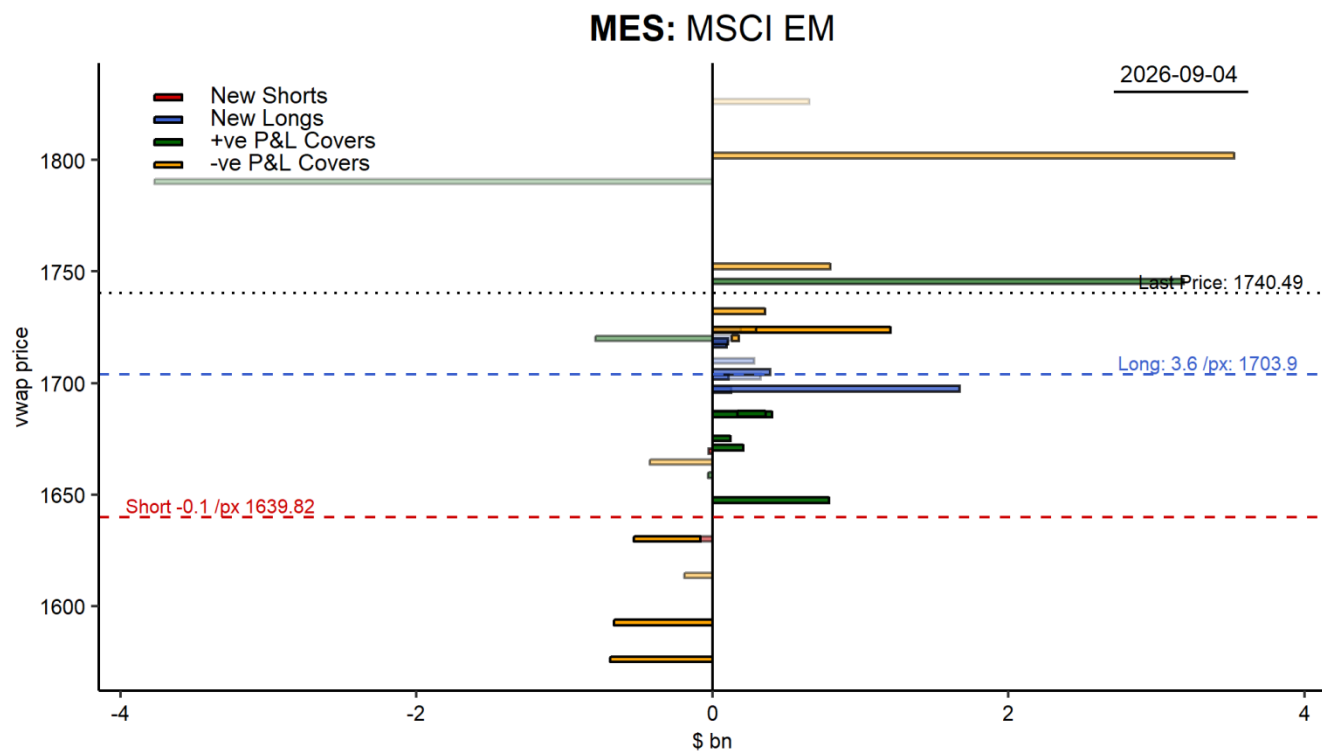


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Source: Citi Research, Bloomberg

MSCI EM

Figure 27. MSCI EM Equity contracts (MES) - position snapshot and time-series

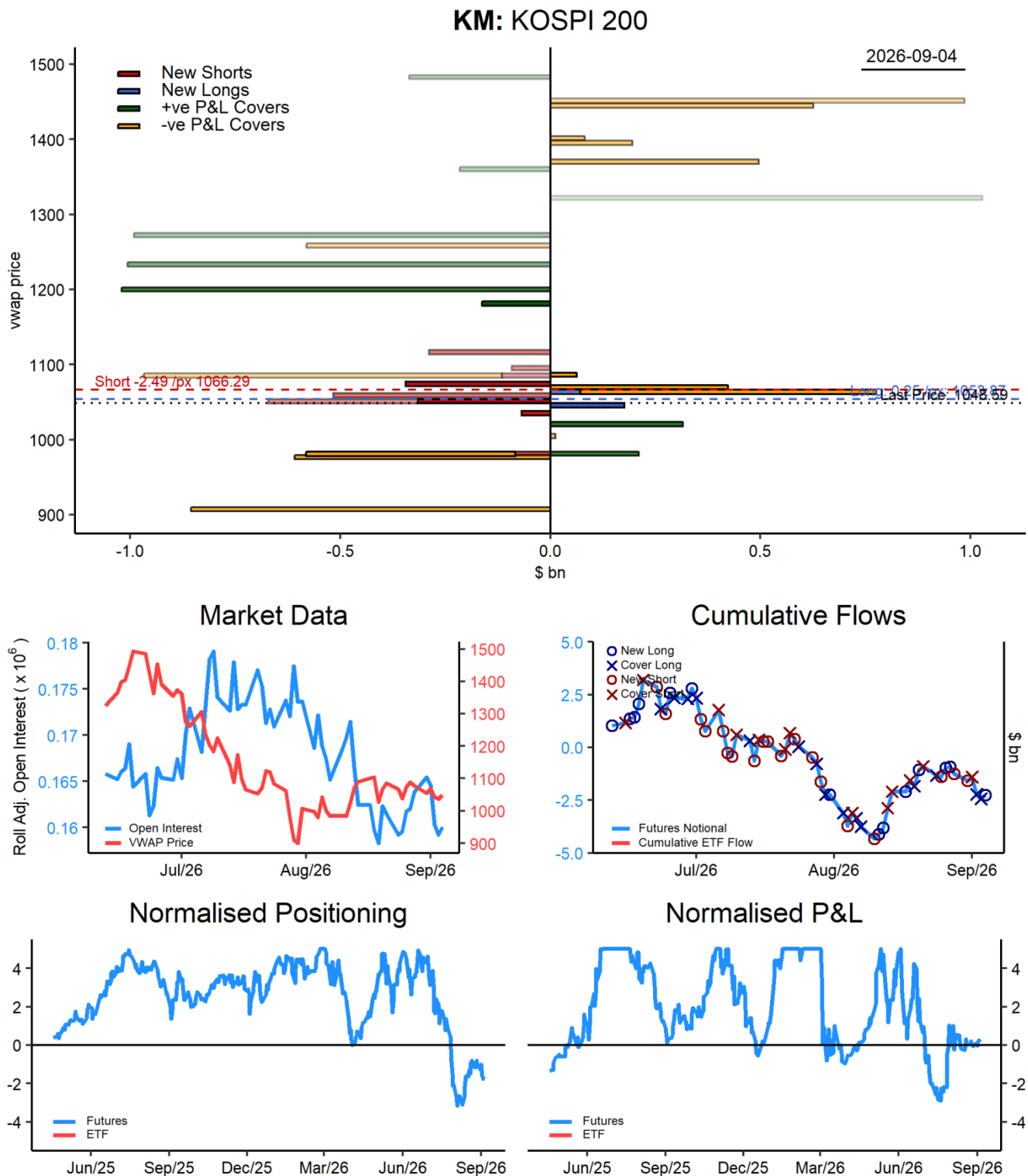


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Source: Citi Research, Bloomberg

KOSPI 200

Figure 28. KOSPI 200 contracts (KM) - position snapshot and time-series

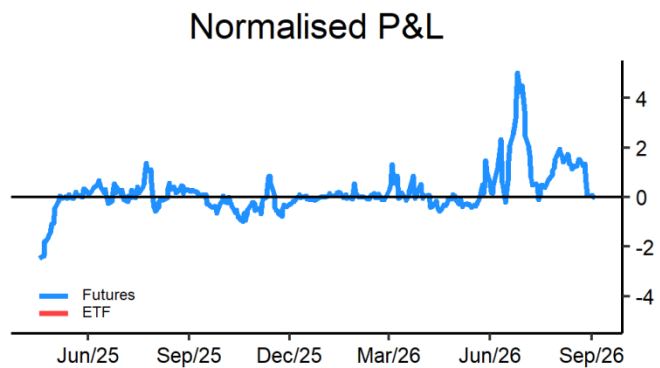
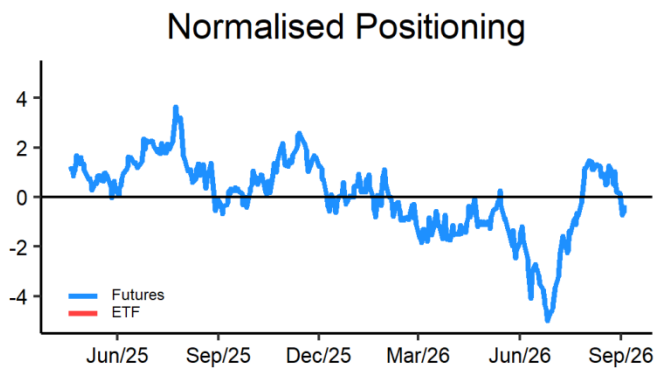
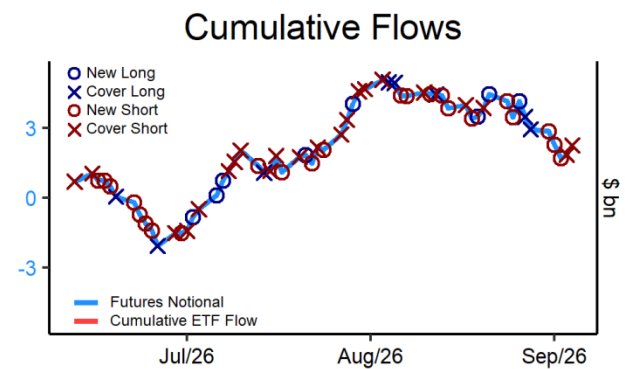
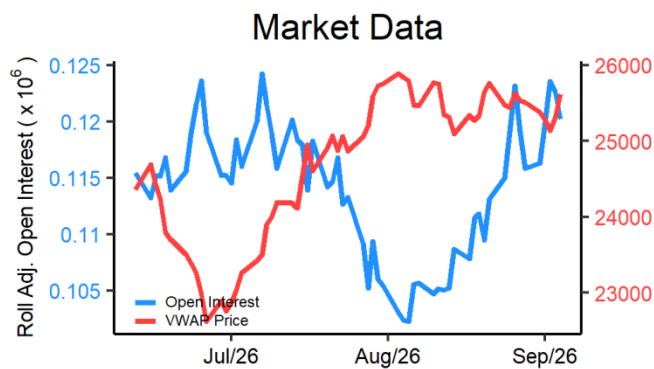
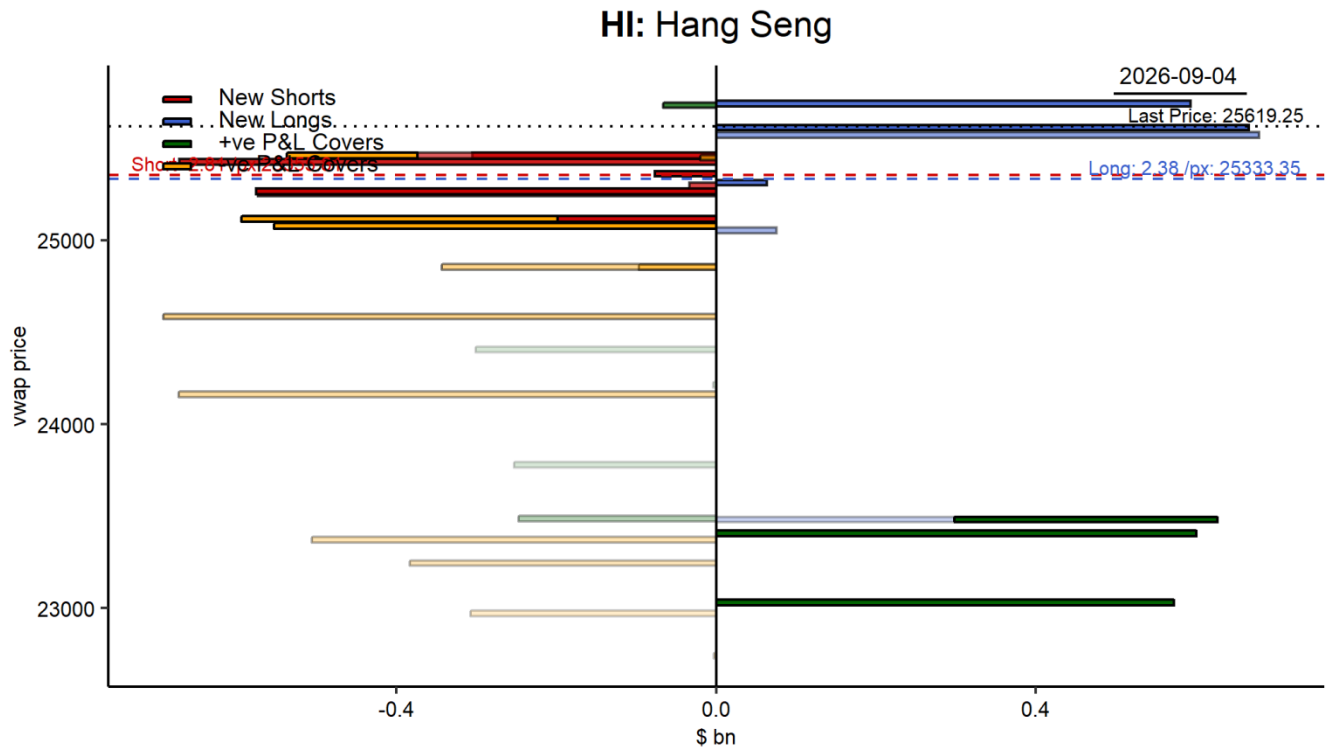


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Source: Citi Research, Bloomberg

Hang Seng

Figure 29. Hang Seng contracts (HI) - position snapshot and time-series

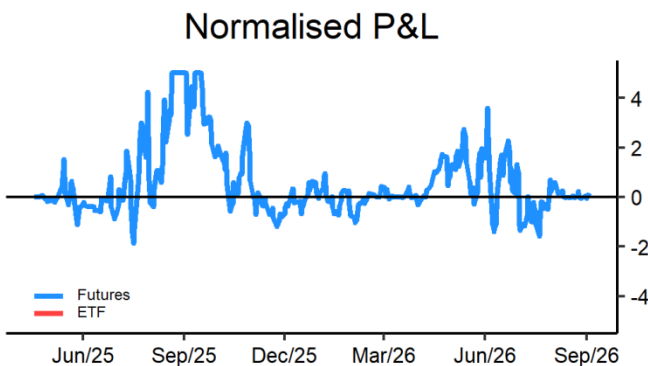
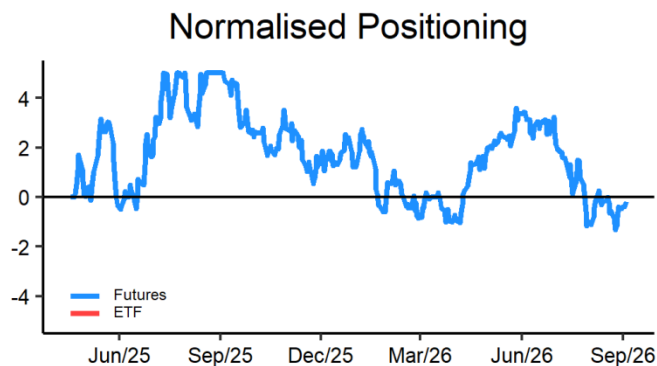
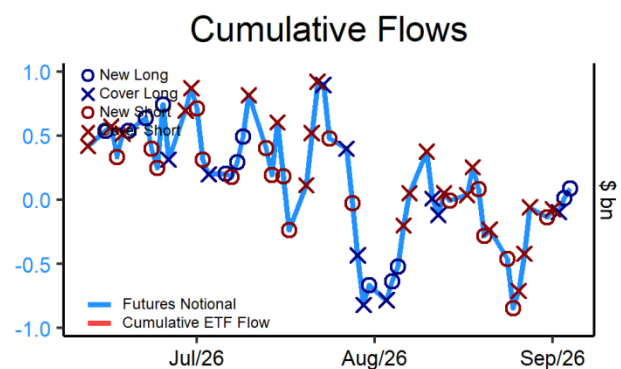
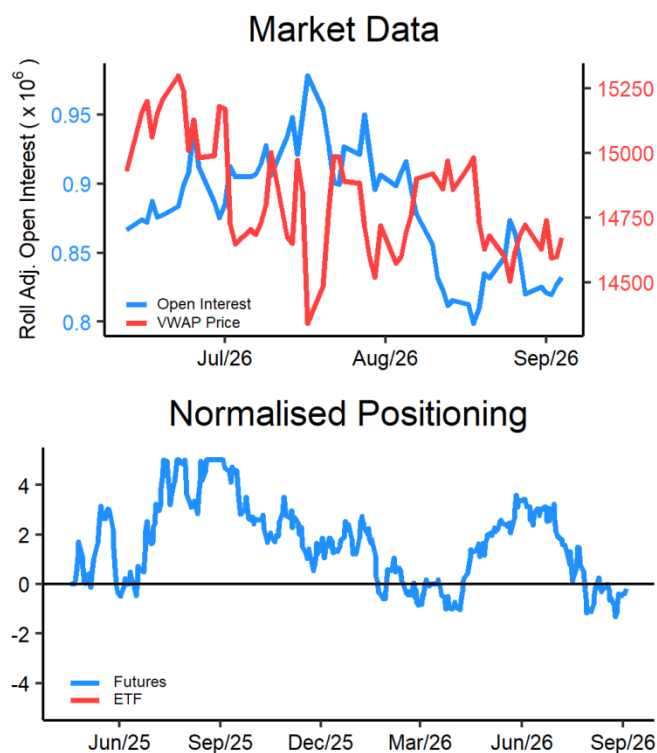
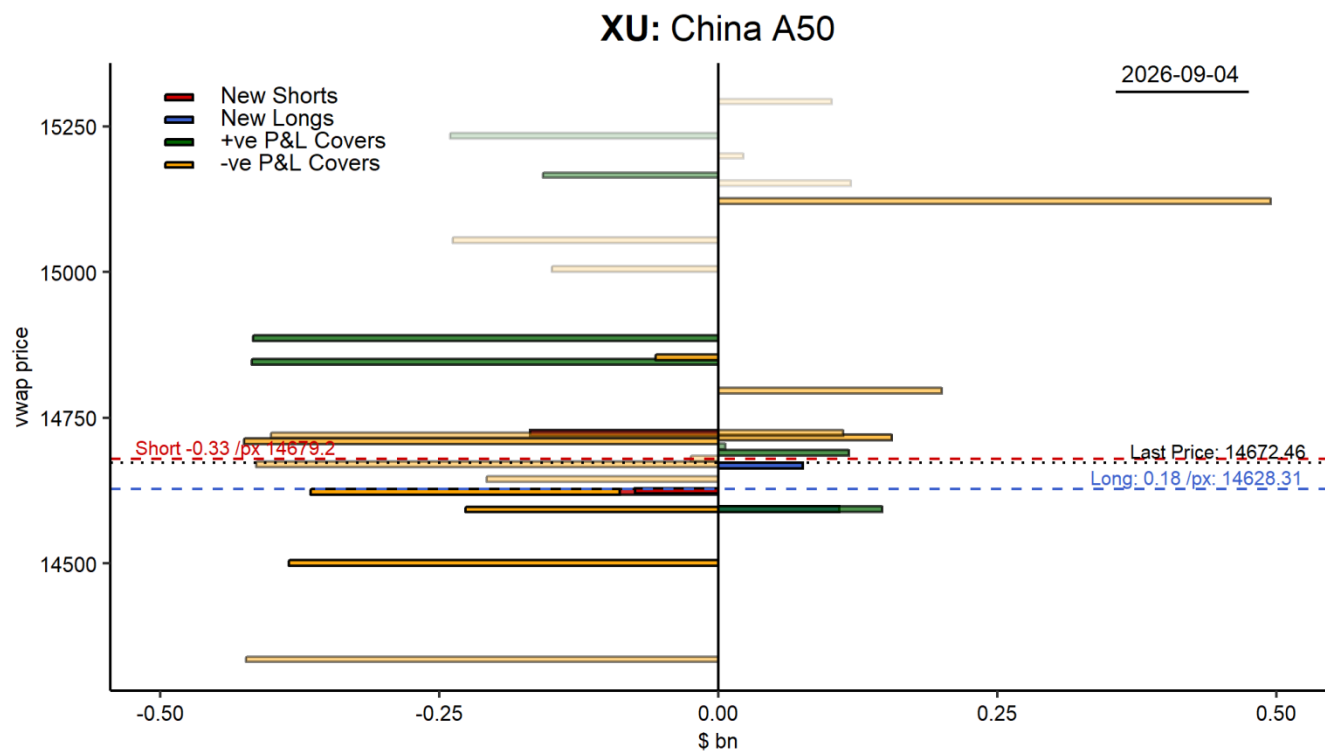


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Source: Citi Research, Bloomberg

China A50

Figure 30. A50 contracts (XU) - position snapshot and time-series



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The Equity Markets Positioning Model – Methodology

The aim of the Equity Markets Positioning Model (EMP) is to provide information on current equity market positioning and how this might influence subsequent price action. It allows readers to ‘visualize’ the investor exposures in equity futures across multiple markets in order to:

- Quantify the current market positioning and profitability;
- Identify extreme market positions;
- Categorize periods in which price action will be dominated by positioning;

The model calculates a ‘headline’ positioning score between -5 to 5 and provides a visualization of the current and past market.

How does the model work?

At a micro scale, price action is dominated by the size, level and speed at which investors choose to execute trades with other financial intermediaries. Buy/sell orders can clearly drive prices higher or lower and the EMP attempts to quantify these flows in order to understand the market direction and motivation.

This is a grand ambition, since there is no one market participant who holds a complete view of all market flows and we are only analyzing and aggregating futures trading to extrapolate the positioning across the whole market (i.e. including all cash trading). Based on this information we estimate:

- Market positioning: What is the outstanding market positioning (i.e. has there been net buying or selling recent history)? At what price and size were these trades executed?
- Market P&L: What is the unrealized profit or loss within the market (given the current outstanding market positioning)? When is this a motivation for investors to risk manage their positions?

We recognize that there are limits to this analysis – incomplete trade visibility, opaque instruments and unknown investor trading strategies create uncertainty in the ‘true’ picture of high-level equity market positioning.

The EMP model calculates the historical equity market positioning (over the last 3mths) and the associated unrealized P&L for the main markets around the world. A breakdown of the assets and contracts used in the EMP are presented in Figure 31 and calculation methodologies are outlined below.

Figure 31. Equity Markets covered by EMP Model

Country/Region	Index	Futures Ticker	ETF Flows
US	S&P 500	ES	Yes
US Tech	Nasdaq 100	NQ	Yes
US SMID	Russell 2000	RTY	No
Europe	Eurostoxx 50	VG	Yes
Germany	DAX 40	GX	Yes
U.K.	FTSE 100	Z	Yes
European Banks	SX7E	CA	Yes
Australia	S&P ASX 200	XP	No
Japan	Nikkei 225	NI	Yes
EAFE	MSCI ex-US	MFS	No
Emerging Markets	MSCI EM	MES	Yes
Korea	KOSPI 200	KM	No
Hong Kong	Hang Seng	HI	No
China	A50	XU	No

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Source: Citi Research

Positioning Calculations

The model identifies an objective subset of positions entered into (or closed) on a given day and classify these as being the relevant marginal long or short positions that determined (being the 'aggressor') the price move that day. In the report and in our analysis we will simply refer to these positions as longs, shorts and net position but they will always be a reference to this concept of a marginal positioning. This presumption then is that this subset of marginal positions are the most price sensitive, i.e. more likely to generate new flows (e.g. profit taking or squeezes) in response to future price moves.

In practice the model uses public trade activity data on the exchange traded futures contracts which is available daily. Naturally, net positioning is always zero for s, since the exchange matches buyers and sellers and creates and cancels contracts as necessary. However, by tracking the daily changes in overall open interest (the number of contracts after the exchange create or cancels contracts) and combining that measure with the change in price we can make a classification for the marginal positioning change on that day as set out in Figure 32.

The change in open interest is taken to be the relevant marginal subset of positions that day and together with the change in price we determine whether to consider it a change in long or short positions.

Figure 32. Positioning Matrix for Futures Markets

		Open Interest	
		Increasing	Decreasing
Daily Prices	Increasing	New Longs Rising prices and increasing open interest implies the 'aggressor' is the buyer pushing the market higher	Cover Shorts Rising prices and decreasing open interest implies the 'aggressor' is covering shorts (i.e. buying into the rally)
	Decreasing	New Shorts Falling prices and increasing open interest implies the 'aggressor' is the seller pushing the market lower	Cover Longs Falling prices and decreasing open interest implies the 'aggressor' is covering longs (i.e. selling into the sell-off)

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Source: Citi Research

Performing this analysis on a day-by-day basis creates a ‘trade activity’ history over a three-month period which can be converted into a snapshot of cumulative positioning and unrealized P&L. To illustrate the methodology, it is useful to look at a worked example. In Figure 33, we present these calculations for S&P 500 futures in early March 2020.

In this period, open interest increased and prices fell, suggesting an increase in the short investor base. The model shows numerous days where new short positions have been opened and relatively few where longs positions have been established, in addition to covering for some long and short positions. At the end of the period the cumulative uncovered positions and P&L is a simple the sum of the outstanding positions / P&L (see “total” in Figure 33 – cumulative positions equate to -459,041 contracts and 868mIn of unrealized P&L).

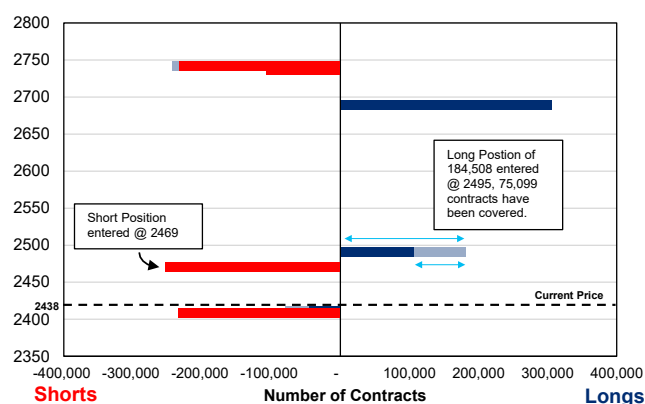
This positioning can be visualized as a horizontal bar graph ordered by price (see Figure 34). It can be clearly seen that the shorts were centered around 2740-2750 and the positions were significantly in the money. This contrasts with the big long position at 2696 which with a falling market was losing position. This snapshot of positioning can be summarized by the cumulative position unrealized P&L levels.

Figure 33. Example of positioning for S&P500 during March 2020

Date	Trade Price	Open Interest	Delta Price	Delta Interest	Trade Activity	Contract Buy/Sell	Cum. Pos.	P&L Snapshot (\$ mil)
06-Mar	2964.0	3,213,000						
09-Mar	2747.8	3,442,251	-7.3%	229,251	New Short	-229,251	-229,251	3,551
10-Mar	2865.8	3,438,501	4.3%	-3,750	Cover Short	3,750	-225,501	-58
11-Mar	2740.3	3,549,717	-4.4%	111,216	New Short	-111,216	-336,717	1,681
12-Mar	2469.0	3,808,784	-9.9%	259,067	New Short	-259,067	-595,784	402
13-Mar	2696.0	4,124,586	9.2%	315,802	New Long	315,802	-279,982	-4,074
16-Mar	2416.3	4,369,586	-10.4%	245,000	New Short	-245,000	-524,982	-266
17-Mar	2495.5	4,554,094	3.3%	184,508	New Long	184,508	-340,474	-531
18-Mar	2414.0	4,628,421	-3.3%	74,327	New Short	-74,327	-414,801	-89
19-Mar	2403.3	4,553,322	-0.4%	-75,099	Cover Long	-75,099	-489,900	216
20-Mar	2438.0	4,522,463	1.4%	-30,859	Cover Short	30,859	-459,041	37
Total							-459,041	868

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Source: Citi Research

Figure 34. Example visualization of positioning



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In EMP, the daily volume weighted average price (VWAP) is used in the calculations (rather than closing prices), since it creates more robust results. In addition, the EMP model chooses to cover trades that have the maximum economic impact (i.e. trades with the largest gains or losses are closed out first). However, in every other aspect, the methodology is identical to the one presented above.

Finally, the reported numbers in the EMP model are normalised into a range from -5 to +5 (representing large shorts / losses to large longs / profits) in order to compare across asset / instruments.

What output is generated by EMP?

The EMP model provides a snapshot of the current positioning (see Figure 35 to Figure 37). These tables summarize the outstanding exposure and unrealized P&L (for a 3mth period in both futures and ETFs).

Typical output is shown in Figure 35 for cross equity market positioning in futures. In this illustration, investors are currently neutral to net short Eurostoxx 50 contracts, which has decreased from a net long position of the past 3 months. This is in contrast to MSCI EM futures where investors are sitting on profitable positions.

Figure 35. Example of Equity Market Positioning and PnL

All metrics in table are normalised between +5 and -5

Investors currently net long S&P 500 Emini futures

Investors currently neutral to net short EuroStoxx 50

(\$bn) 03/09/2020		S&P 500	DAX	FTSE 100	ESToxx50	Nasdaq	Nikkei	MSCI World x-US	MSCI EM
		ES	GX	Z	VG	NQ	NI	MFS	MES
Normalised Scores (-5 / +5)									
Positioning	Future	1.7	-0.1	1.0	-0.0	4.8	1.1	-0.7	2.0
	ETF	3.0	1.4	-0.6	0.9	3.6	1.0		1.1
	3m Net Pos	2.3	0.7	0.2	0.4	4.2	1.1	-0.7	1.6
	1w Change	0.4	0.2	0.3	0.3	-0.3	0.3	-0.4	-0.3
P&L	Future	1.2	-0.1	-0.7	-0.0	4.8	0.5	0.1	1.8
	ETF	3.7	1.9	-0.5	1.1	3.9	1.3		1.4
	3m Net PnL	2.5	0.9	-0.6	0.5	4.4	0.9	0.1	1.6
	1w Change	0.3	0.3	-0.0	0.3	-0.4	0.3	0.0	-1.1

Investors are sitting on profitable positions in MSCI EM.

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Source: Citi Research

In addition, we further breakdown market positioning by analyzing both long and short positions, the average entry price and the position relative to trading over the past three months (Figure 36).

Figure 36. Example of Breakdown of Market Positioning – Not Current

Net Notional = Longs minus Shorts (in \$blns)

Percentile refers to positioning relative to past 3 years

% Offside refers to what percentage of shorts are below Ref price – indicates risk of short squeeze if market rallies. Opposite for Longs

		S&P 500	DAX	FTSE 100	ESTOxx50	Nasdaq	Nikkei	MSCI World x-US	MSCI EM
(\$bn)	03/09/2020	ES	GX	Z	VG	NQ	NI	MFS	MES
Futures	Net Notional	31.7	-0.1	2.4	-0.4	10.8	1.2	-1.1	6.4
	Percentile	67%	36%	53%	33%	99%	64%	11%	75%
Shorts	Notional	-7.5	-1.3	-3.9	-1.4	-1.6	-1.1	-2.7	-0.7
	Percentile	24%	37%	81%	24%	28%	20%	80%	21%
	Avg. Entry	3,246	12,994	6,116	3,316	10,993	22,959	1,890	1,089
	% offside	100%	100%	3%	100%	100%	100%	32%	89%
Longs	Notional	39.2	1.2	6.3	0.9	12.4	2.3	1.6	7.0
	Percentile	57%	18%	72%	10%	99%	25%	18%	67%
	Avg. Entry	3,337	13,068	6,096	3,311	11,166	22,889	1,893	1,051
	% offside	13%	0%	100%	0%	18%	0%	60%	38%
	Ref Price	3,463	13,304	5,909	3,366	11,803	23,490	1,878	1,094
Activity	1-day	-2.6	-0.2	0.0	-0.1	-0.9	0.3	0.0	0.0
	Type	longs	shorts	longs	shorts	longs	longs	shorts	shorts
	1-week	4.2	0.1	0.1	-0.5	-0.4	0.1	-0.7	0.5

Average entry price over past 3 months

1 day trading activity in \$blns. Negative (positive) means risks is being cut (added)

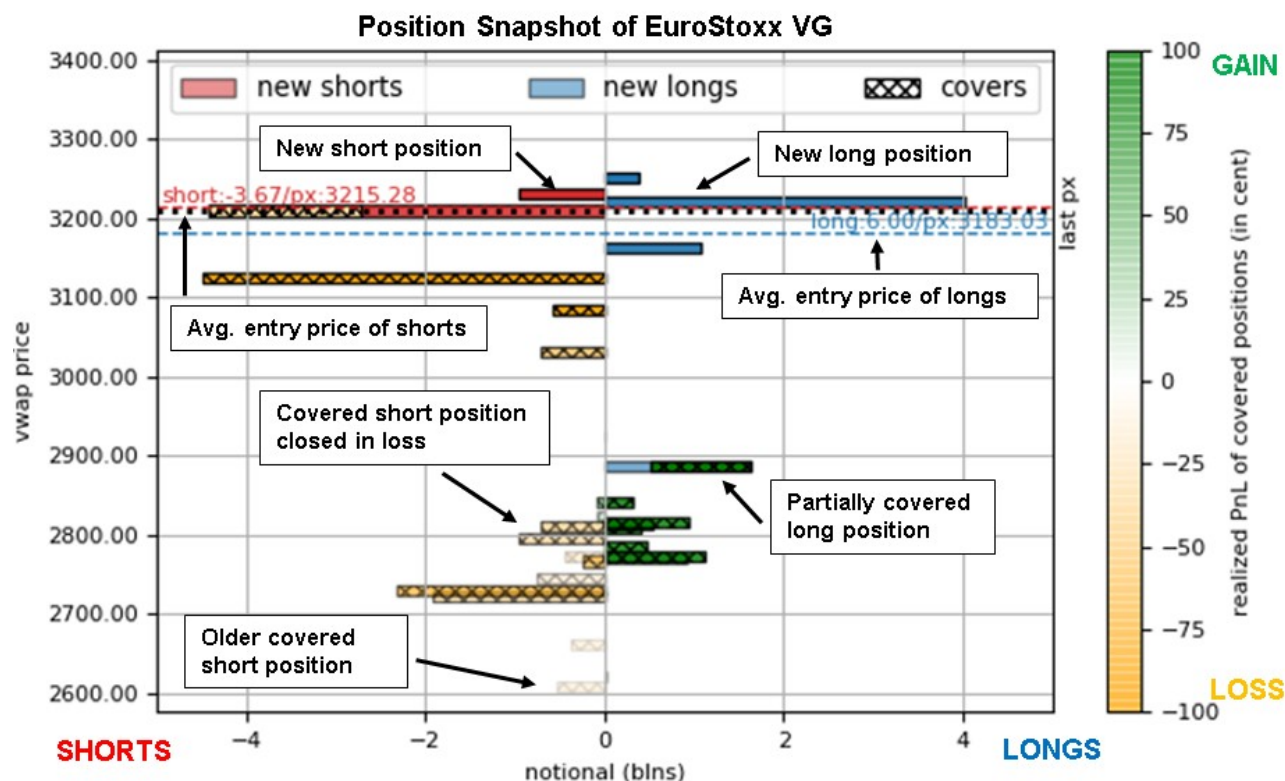
Defines what side the risk is being traded

1 week cumulative trading flow

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Source: Citi Research

Lastly, a complete visualization of the positioning is presented in Figure 37. This shows the current positioning ordered by VWAP prices over a 3-month period. The bars in the chart illustrate the level at which trades were executed (red = short positions and blue = long positions). The lighter bars imply older trade activity. When a trade is covered the bar is hashed with XXX and the color of the shading represents if it was covered at a gain or loss (yellow=loss, green=gain). The horizontal red and blue dotted lines show the respective average entry price of contracts over the past three months.

Figure 37. Example Snapshot of Position in EuroStoxx Futures – Not Current



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Source: Citi Research

What it means for Cash Equities

Although traditionally used as a hedge, futures are also used by investors for taking an outright view on markets. This becomes increasingly relevant when futures positioning is extended. We explore this content in greater detail in [Regime Modelling using Futures Positioning: Futures vs. Cash – A Distant but Meaningful Relationship](#), which explores the interplay between futures positioning and behavior seen in underlying cash markets.

Previously Published Research

Figure 38. Research Reports (since 2009)

Date	Research Theme	Report Link
20-Oct-25	Allocation	Searching for Alpha: What can Mutual Fund Flows tell us about Future Stock and Sector Returns?
29-Sep-25	Factor	What Works in Equity Markets: Misguided by Factor Betas
04-Sep-25	Rotation	What Works in Equity Markets: Style Factor Performance and High Macro Risk Environments
01-May-25	Positioning	What Works in Equity Markets: Does Crowding Outperformance Outweigh Crash Risk?
23-Apr-25	Rotation	Searching for Alpha – Style Rotation: Using Mutual Fund Flows to Predict Style Returns
05-Dec-24	Factor	Searching for Alpha – Stock Selection and Enhanced Growth Strategies with Option-Implied Volatility
29-Oct-24	Allocation	Searching for Alpha – Macro Sector Selection – Using Macro Regimes & News Sentiment to Forecast Sector Returns
27-Aug-24	Positioning	Measuring the Crowded Trade in Global Equities News Sentiment, Factor Crowding & Extended Coverage
20-Aug-24	Misc.	Global Risk Attribute Model: A Framework for Equity Macro Risk Analysis
10-Apr-24	Misc	Searching for Alpha – European Industry-Specific Return Drivers
11-Jan-24	Allocation	Searching for Alpha – Are PMIs Leading or Misleading?
04-Dec-23	Factor	Searching for Alpha: Leveraging Option Information to Improve Factor Performance
26-Sep-23	Allocation	Searching for Alpha: Macro Regimes and Systematic Equity Investing
10-Aug-23	Allocation	Searching for Alpha: Exposure to Macro via Equities
26-Jun-23	Positioning	Measuring the Crowded Trade in Global Equities: Introducing our Short Stock Crowding Model
16-May-23	Machine Learning	Searching for Alpha: From Machine Learning to Stock Screens – A Rulefit Model
24-Mar-23	Factor	Searching for Alpha: A Bottom-Up Approach to Evaluating Market Overbought/Oversold Conditions
20-Jan-23	Factor	What Works in Equity Markets: The Enduring Appeal of Income Investing
13-Jan-23	Factor	What Works in Equity Markets: Return on Equity, a different perspective through Revisions
07-Nov-22	Event	Searching for Alpha: Systematic M&A Strategies that Outperform Discretionary Event-Driven Approaches
26-Oct-22	Allocation	What Works in Equity Markets: Sector Selection – A Bottom-up, Factor Approach
09-Jun-22	Event	Searching for Alpha: M&A – Identifying Potential Takeover Targets
24-Feb-22	Event	What Works in Equity Markets: Buy Backs – Back in Vogue
26-Jul-21	Factor	What Works in Equity Markets: Risky Value vs. Quality Value Revisited
26-Jul-21	Machine Learning	Searching for Alpha: Practical Considerations in Building Neural Networks for Stock Selection
22-Jul-21	ESG	ESG Insights: Indexed ESG Funds – A Widespread Implementation Choice
22-Jun-21	Factor	Searching for Alpha: Go with the Stock Connect Flow, for a Sharpe of 4
18-Jun-21	Factor	Searching for Alpha: Overlaying Price Mo. and Short Interest to avoid drawdowns and improve returns
15-Jun-21	ESG	ESG Insights: ESG Fundamental Characteristics
13-Apr-21	ESG	ESG Insights: Following the ESG Herd: Has ESG become a Crowded Trade?
29-Mar-21	Machine Learning	Searching for Alpha: Pairs Trading: Applying Machine Learning to Pairs Trading
09-Mar-21	Positioning	Regime Modelling using Futures Positioning: Futures vs. Cash – A Distant but Meaningful Relationship
18-Feb-21	ESG	ESG Insights: The ESG ‘Premium’: How does ESG compare to other Factors?
01-Feb-21	Event	Searching for Alpha: Asia Earnings Surprise: Predicting Asia Earnings Surprises
14-Jan-21	ESG	ESG Insights: Where Machines do it Better: Inferred ESG Ratings Data
30-Oct-20	Factor	Searching for Alpha: China A Alpha – Smoothing Price Momentum
16-Oct-20	Factor	Under the Microscope: Overlapping Momentum
14-Sep-20	Positioning	Equity Markets Positioning Model: Introducing the EMP
11-Sep-20	Factor	Searching for Alpha: Combining ESG and Risk Premia Investment: A Double Screen Approach
08-Sep-20	Factor	Searching for Alpha: Factoring Short Interest: Measuring and Profiting from Information in Shorts
28-May-20	Machine Learning	Searching for Alpha: Machine Learning – SHAP Value as Factor Selection Criterion
28-Feb-20	Machine Learning	Searching for Alpha: Machine Learning: Leveraging Return Forecasts
21-Feb-20	Factor	Searching for Alpha: China A Alpha – Sector-rel. Value Is Powerful, Choose Sector Mappings Wisely
03-Oct-19	Machine Learning	Searching for Alpha: Machine Learning: Beyond Random Forest for Stock Selection
23-Sep-19	Event	Searching for Alpha: Earnings Surprise: Managing Expectations
05-Mar-19	Machine Learning	Searching For Alpha: Machine Learning: Interpreting Machine Learning Predictions
07-Feb-19	Misc	Searching for Alpha: GAAP vs. Non-GAAP: Which Earnings does the Market Price?
21-Nov-18	Positioning	Searching for Alpha: Style Crowding in Asia: Getting Ahead of the Crowd
17-Sep-18	Event	Searching for Alpha: Earnings Surprise: Using ML to Forecast Earnings Surprises & Returns
13-Sep-18	ESG	Searching for Alpha: The ESG Edge: ESG Investing – A Step Forward
03-Sep-18	Machine Learning	Searching For Alpha: Machine Learning: Interacting Machine Learning and Factors
03-May-18	Positioning	Measuring the Crowded Trade: Introduction to our Crowding Composite for individual stocks
14-Mar-18	Factor	Searching for Alpha: Profiting from Capex: Look To Capex Announcements
07-Sep-17	Rotation	Searching for Alpha: Tactical Style Rotation: Using Risk and Return to Manage Style Exposure
10-Mar-17	Misc	Searching for Alpha: Big Data: Navigating New Alternative Datasets
13-Feb-17	Factor	Searching for Alpha: Betting Against (Accurate) Beta
19-Sep-16	Factor	Searching for Alpha: Competitive Advantage: Survival of the Fittest
09-Sep-16	Factor	Searching for Alpha: Financial Strength Redux
14-Apr-16	Rotation	Searching for Alpha: Dynamic Style Weighting: Risk-Based Equity Style Allocation
04-Mar-16	Smart Beta	Long-Only Pure Style Portfolios: No Shorts Please
18-Feb-16	Factor	Industry Alpha Insights: Banks: One Size Does Not Fit All
21-Sep-15	Factor	Under the Microscope: Stock Momentum Conflation
15-Sep-15	Allocation	Searching for Alpha: Macro Moves Markets: Economic Data, Expectations and Market Pricing
06-Aug-15	Misc	Normalising for Style Factor Composites: Reviewing alternative solutions for constructing style composites
23-Mar-15	Rotation	Searching for Alpha: Style Timing: Style Performance, Trading Volumes and Investor Agreement
13-Mar-15	Misc	World Radar Screen: Refining Our Global Search for Alpha
18-Feb-15	Factor	Searching for Alpha: Networking with Analysts: Modelling Analyst Forecast Dependence
01-Oct-14	Smart Beta	The Rise of Low Risk Investing: Is It Getting Crowded Out There?
27-Mar-14	Factor	Under the Microscope: Five Innovations In Momentum Investing
07-Mar-14	Factor	Searching for Alpha: Timing Price Momentum: When Does Momentum Move?
27-Nov-13	Smart Beta	Equity Risk Premia Investing: A New Methodology for Monitoring Style Performance
23-Jul-13	Allocation	Stock Market Country Selection: Changes to a Well-Established Model

02-Jul-13	Factor	Searching For Alpha: Digging For Dividends: QUARI - QUality with A Reliable Income
24-Jun-13	Misc	Global Theme Machine: An Objective Way of Identifying Attractive Investment Themes
25-Mar-13	Factor	Searching for Alpha: Purifying Analyst Recommendations: Removing Beta to get to the Alpha
06-Nov-12	Factor	Searching for Alpha: Tangible Benefits of Intangibles: Brand, Respect & Intellectual Capital
09-Mar-12	Smart Beta	Low-Risk Portfolio Strategies: Sharpe Ratio Maximisation and Multi-Asset Applications
28-Feb-12	Rotation	Macro Risk and Style Rotation: A Guide Rather than a Prescription
14-Sep-11	Factor	Searching for Alpha: Accruals Volatility - A New Approach to Quality Investing
24-Aug-11	Allocation	Industry Alpha Insights: Four Approaches to Tactical Industry Selection
17-Mar-11	Misc	Industry Alpha Insights: Quantifying Industry-Specific Fundamentals
18-Nov-10	Smart Beta	Low-Risk Equity Portfolios: More than just Minimum Variance
15-Nov-10	Allocation	Under the Microscope: Measuring Systemic Risk - The Absorption Ratio
14-Jun-10	Factor	Under the Microscope: Optionality in Valuation
31-Mar-10	Event	Searching for Alpha: Earnings Surprise: Still Profiting from Surprises
29-Jan-10	Factor	Momentum in Japan: Looking at Price, Trading Values and Earnings
15-Oct-09	Rotation	Searching for Alpha: Style Rotation: Optimising Style Rotation Strategies

Source: Citi Research

Citi Quant Research Team

Figure 39. Citi Quantitative Research Teams (For Informational Purposes Only)

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Appendix A-1

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Data current as of 01 Jul 2026	12 Month Rating			Catalyst Watch		
	Buy	Hold	Sell	Buy	Hold	Sell
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% of companies in each rating category that are investment banking clients	38%	42%	27%	42%	37%	36%
Citi Research Quantitative World Radar Screen Model Coverage	30%	40%	30%			
% of companies in each rating category that are investment banking clients	24%	16%	16%			
Citi Research Quantitative Latam Radar Screen Model Coverage	23%	57%	21%			
% of companies in each rating category that are investment banking clients	39%	40%	43%			
Citi Research Quantitative Asia Radar Screen Model Coverage	20%	60%	20%			
% of companies in each rating category that are investment banking clients	26%	10%	7%			
Citi Research Quantitative Australia Radar Screen Model Coverage	41%	0%	59%			

% of companies in each rating category that are investment banking clients	35%	0%	4%
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